AI-BASED AYURVEDA AND SPIRITUAL WELLNESS TOURISM MATCHING FOR SRI LANKA

INDIVIDUAL THESIS

D.C.M. De Silva

(IT22108586)

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DECLARATION AND SUPERVISOR CERTIFICATION

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Student name: D.C.M. De Silva

Student ID: IT22108586

Signature: _____ Date: _____

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Role and name	Signature	Date
Supervisor: Mr Buddika Harshnath		
Co-supervisor: Prof. Samantha Rajapaksa		

ABSTRACT

Visitors seeking Ayurvedic treatment or a meditation retreat in Sri Lanka must choose a centre for a multi-week stay without reliable information on which centres treat their condition, suit their Ayurvedic constitution, fit their budget and are safe. Existing tourism recommenders and booking portals do not consider health condition or constitution, and dosha classification research stops at a label without connecting it to real providers.

This thesis presents the author's individual contribution to the Tour Ceylon platform, the Wellness Component: an explainable, multilingual AI system that matches tourists to 83 curated Sri Lankan Ayurveda centres and spiritual retreats. A Random Forest classifier predicts the tourist's dominant dosha from six questionnaire features, selected from six algorithms with a test accuracy of 0.810, weighted F1 of 0.794 and 5-fold cross-validated accuracy of 84.3%. A transparent weighted model scores centres on condition, dosha, review-derived quality and budget. An ablation study over 60 simulated queries showed that the full model raises the average quality of the top match from 3.41 to 4.61, budget alignment from 50.0% to 90.0% and dosha compatibility from 10.0% to 51.7% compared with condition-only matching. Ten per-language TF-IDF and SVM classifiers interpret free-text symptoms and retreat goals in English, German, Russian, Chinese and Japanese, and a hybrid keyword and machine learning layer screens guest reviews for safety concerns. A regression model estimates treatment outcome with a mean absolute error of 4.7 percentage points.

Implemented with Flask, scikit-learn, PostgreSQL and a five-language web interface, the component passes all 75 automated test executions, returns a recommendation in 22 ms at the median and is integrated with the platform's shared backend and database. The main limitation is reliance on expert-rule variations of six real cases for training, which makes validation with real tourists the priority for future work.

Keywords: wellness tourism, Ayurveda, dosha classification, recommender systems, random forest, multilingual text classification

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
API	Application Programming Interface
CRUD	Create, Read, Update and Delete
CSV	Comma-Separated Values
CV	Cross-Validation
E2E	End-to-End
F1	Harmonic mean of precision and recall
FR	Functional Requirement
GWI	Global Wellness Institute
HTTP	Hypertext Transfer Protocol
JSON	JavaScript Object Notation
LOO	Leave-One-Out
MAE	Mean Absolute Error
ML	Machine Learning
NFR	Non-Functional Requirement
NLP	Natural Language Processing
OSM	OpenStreetMap
OWASP	Open Worldwide Application Security Project
PDPA	Personal Data Protection Act (Sri Lanka, No. 9 of 2022)
REST	Representational State Transfer
RF	Random Forest
RQ	Research Question
SDG	Sustainable Development Goal
SLTDA	Sri Lanka Tourism Development Authority
SUS	System Usability Scale
SVM	Support Vector Machine
TF-IDF	Term Frequency–Inverse Document Frequency
UAT	User Acceptance Testing
UI	User Interface

1 INTRODUCTION

This chapter introduces the research. It first places the work within the Tour Ceylon group project and the wellness tourism domain, then states the problem, the research gap, the aim and objectives and the research questions. It closes with the scope, the author's individual contribution and the structure of the thesis.

1.1 Overall project context

Tourism is one of Sri Lanka's most important sources of foreign exchange. The Sri Lanka Tourism Development Authority (SLTDA) recorded 2,362,521 international arrivals in 2025, a 15.1% increase over 2024 [1]. Visitors increasingly expect digital services that plan their trip, keep them safe and respond to their individual needs, yet most of the information they rely on is scattered across booking portals, review sites and word of mouth.

The group research project **Tour Ceylon (R26-IT-152)** addresses this gap with an AI-powered travel platform for Sri Lanka built by four students. Each member owns one research component that also works as a standalone module, and the components are joined into one platform through a shared Flask backend, a shared PostgreSQL database and a common web front end. The platform covers (i) itinerary and travel planning, (ii) tourist safety and emergency (SOS) support, (iii) tourism risk management and live data analysis, and (iv) wellness tourism, which is the component reported in this thesis.

This thesis covers only the wellness component. The other components are mentioned only where the wellness component exchanges data with them or depends on their shared services (Section 5.3).

1.2 Individual component context

Wellness tourism is travel motivated by the wish to maintain or improve personal wellbeing. The Global Wellness Institute (GWI) valued the global wellness tourism market at USD 830.2 billion in 2023 and projects USD 1,351.0 billion by 2028 [2]. Sri Lanka's traditional Ayurveda and its Buddhist meditation heritage give the country a distinctive position in this market [3], [4]. SLTDA statistics show that

12,977 visitors in 2025 stated health or Ayurveda as their main purpose of visit, that Germany was the largest source market for this purpose (6,363 visitors), and that these visitors stayed an average of 14.7 days, the longest of any purpose [1].

Choosing an Ayurveda centre or a meditation retreat is, however, a high-stakes decision for a foreign visitor. Treatment programmes run for two to five weeks, the quality of centres varies widely, and the information needed to choose well (which centres treat a given condition, whether a centre suits the visitor's constitution, what previous guests experienced, and what the stay will cost) is not available in one place or in the visitor's language. Local practitioners themselves report weak regulation, inconsistent quality and poor information sharing as barriers to the sector [5].

The **Wellness Component** developed in this work is an AI-based matching system that recommends Ayurveda treatment centres and spiritual or meditation retreats in Sri Lanka. It profiles the tourist's Ayurvedic constitution (dosha) with a machine learning (ML) classifier, matches the tourist's condition, constitution and budget against a curated database of 83 real centres, scores centres with quality signals derived from real guest reviews, screens reviews for safety concerns, and presents explainable results, cost estimates and practical guidance in five languages.

1.3 Problem statement

International visitors who want Ayurvedic treatment or a meditation retreat in Sri Lanka have no reliable way to find centres that suit their health condition, constitution, budget and language, and no structured way to judge the quality and safety of those centres before they travel.

Four specific problems follow from this statement:

1. **Generic search.** Booking portals and map services list centres by location and star rating. They do not take the visitor's condition or Ayurvedic constitution into account, although constitution is central to how Ayurveda personalises treatment [6], [7].
2. **Inaccessible expertise.** Determining a person's dosha normally requires a consultation with a practitioner. Online dosha quizzes exist [8], but their output is not connected to any provider, so the visitor still has to search manually.

3. **Unstructured quality and safety information.** Guest reviews contain valuable signals about hygiene, professionalism and safety, but they are free text spread over many pages and are rarely summarised [9].
4. **Language and cost barriers.** The largest wellness source markets (Germany, Russia, China, Japan) are not served in their own languages, and visitors cannot easily estimate the total cost of a multi-week programme.

1.4 Research gap

Existing work falls into three groups that do not meet (Section 2.5). Tourism recommender systems rank attractions and accommodation using user preferences and context, but they do not model health condition or constitution [10]. Health-tourism recommenders match patients to providers by service and location, again without constitution-aware personalisation [11]. Research on Ayurvedic constitution shows that prakriti and dosha can be described objectively and even predicted with ML from questionnaires [12], [13], [14], but this work stops at classification and does not connect the result to a real provider.

No published system or product identified in this review combines (a) ML-based dosha profiling from a short tourist-friendly questionnaire, (b) condition- and budget-aware matching against a verified database of real Sri Lankan providers, (c) review-derived quality and safety signals and (d) multilingual, explainable output. The gap is particularly clear for Sri Lanka, where the sector itself calls for better information sharing [5]. Closing this gap is the research problem of this component.

1.5 Aim and objectives

Aim. To design, implement and evaluate an explainable, multilingual AI system that matches international wellness tourists to suitable Ayurveda treatment centres and spiritual retreats in Sri Lanka using their condition, Ayurvedic constitution, budget and review-derived quality and safety signals.

Objectives.

1. **O1.** Build a curated, verified dataset of real Ayurveda and meditation centres in Sri Lanka, enriched with condition coverage, price tier, location and guest reviews.
2. **O2.** Train and select a classifier that predicts a tourist's dominant dosha from a short lifestyle questionnaire, and justify the choice against alternative algorithms.
3. **O3.** Design a transparent weighted matching model that combines condition, dosha, quality and budget, and measure the contribution of each factor with an ablation study.
4. **O4.** Provide multilingual access through per-language free-text symptom classifiers and a five-language interface for the main source markets.
5. **O5.** Screen guest reviews for safety concerns with a hybrid keyword and ML approach, and estimate treatment duration, outcome and cost.
6. **O6.** Deliver the component as a tested, secure full-stack application that integrates with the Tour Ceylon platform, and evaluate it for correctness, performance and usability.

1.6 Research questions or hypotheses

The component addresses four research questions (RQ):

- **RQ1.** Can a supervised classifier predict a tourist's dominant dosha from six questionnaire features with useful accuracy, and which algorithm performs best on this data?
- **RQ2.** Does adding dosha compatibility, review-derived quality and budget fit to condition matching produce better recommendations than condition-only matching?
- **RQ3.** Can lightweight per-language text classifiers map a tourist's free-text description of symptoms or retreat goals to the correct category in five languages?
- **RQ4.** Can a hybrid of keyword rules and a trained classifier identify safety-relevant guest reviews, and what are the limits of each method?

RQ2 is tested as a hypothesis: **H1** – the full weighted model selects top recommendations with higher review quality and better budget and dosha alignment than condition-only matching.

1.7 Scope limitations and assumptions

In scope. Recommendation of Ayurveda centres (curative and wellness or rejuvenative) and spiritual or meditation retreats in Sri Lanka; dosha prediction from a questionnaire; multilingual symptom and retreat-goal entry in English, German, Russian, Chinese and Japanese; review-based quality and safety signals; duration, outcome and cost estimates; an administration dashboard; and integration with the Tour Ceylon platform.

Out of scope. Medical diagnosis or treatment advice, online payment and booking transactions, real-time availability, and clinical validation of treatment outcomes. The system supports a visitor's choice of centre; it does not replace a consultation with a qualified Ayurvedic physician, and the interface states this explicitly.

Assumptions and constraints. (i) The dosha questionnaire captures the current state of imbalance (vikruti) rather than the birth constitution (prakriti). (ii) Real labelled data is scarce: the dosha and outcome datasets are anchored on six real patient cases from the project's domain expert and extended with expert-rule variations (Section 3.2). (iii) Centre information was correct when collected and is maintained through the admin dashboard. (iv) The component must run on an ordinary laptop without a GPU.

1.8 Individual contribution

All work described in this thesis was carried out by the author as the owner of the wellness component. The individual contribution consists of:

- the curated database of 83 real centres with verification notes, 256 guest reviews and review-derived quality scores;
- the dosha classification pipeline, including the six-algorithm comparison and the selected Random Forest model;

- the weighted matching engine with path separation (Ayurveda versus spiritual), companion matching and a per-factor score breakdown, together with the ablation study that evaluates it;
- ten per-language text classifiers (five languages × two domains) and the hybrid review-safety layer;
- the treatment outcome predictor and the duration and cost estimation logic;
- the Flask REST API, the five-language web interface, the admin dashboard and the PostgreSQL schema with JSON fallback;
- a reproducible training notebook, an automated test suite of 48 test cases (75 executions) and a performance benchmark; and
- the integration of the component with the shared Tour Ceylon backend and database.

1.9 Thesis structure

Chapter 2 reviews the literature and existing products and derives the research gap. Chapter 3 describes the research methodology, data and evaluation metrics. Chapter 4 presents the requirements and the design of the component. Chapter 5 explains the implementation, integration and testing. Chapter 6 reports the results and validation, and Chapter 7 discusses them, including limitations. Chapter 8 examines commercialisation and sustainability, and Chapter 9 concludes the thesis and outlines future work. The appendices contain extended results and the academic integrity and AI-use disclosure.

2 LITERATURE REVIEW AND RELATED WORK

This chapter reviews the theoretical foundations and the related research that informed the wellness component, compares existing products and techniques, and closes with a synthesis of the research gap and the conceptual framework of the component.

2.1 Review method and scope

The review followed a structured search of four source types: (i) peer-reviewed research on recommender systems, tourism and health tourism; (ii) research on Ayurvedic constitution and its objective or computational assessment; (iii) research on wellness and Ayurveda tourism in Sri Lanka; and (iv) official statistics and existing commercial products. Searches used Google Scholar, IEEE Xplore, ScienceDirect, SpringerLink, arXiv and official SLTDA and GWI publications with combinations of the terms *wellness tourism*, *Ayurveda tourism Sri Lanka*, *prakriti*, *dosha*, *machine learning*, *tourism recommender system*, *health tourism recommender* and *online reviews quality*.

Sources were included if they (a) addressed recommendation or decision support in tourism or health tourism, (b) studied dosha or prakriti assessment, or (c) provided evidence about the Sri Lankan wellness tourism market. Foundational methods papers were included for every algorithm used in the component. Commercial products were examined directly through their public websites. The review emphasises what each source does *not* cover, because that defines the gap the component addresses.

2.2 Domain and theoretical foundations

2.2.1 Wellness and Ayurveda tourism

Smith and Puczkó distinguish medical tourism, which treats illness, from wellness tourism, which maintains or improves wellbeing, and place Ayurveda in an overlap between the two: it offers both curative programmes and rejuvenation [4]. The World Health Organization's traditional medicine strategy encourages countries to integrate traditional medicine safely and to regulate practitioners and products [15]. In Sri

Lanka, Fonseka and Pavithra surveyed 50 wellness tourists and found that 34% visited specifically for Ayurveda and holistic healing, and that the range of Ayurveda and holistic treatments was the most influential destination capability, explaining 42.6% of the variance in their factor analysis [3]. The same study identified a lack of standardisation and certification as a key weakness of the sector. Dharmapala and Gnanapala interviewed 21 Ayurvedic practitioners and reported weak government support, a shortage of qualified staff, inconsistent regulation and poor information sharing [5].

2.2.2 Ayurvedic constitution (dosha and prakriti)

Ayurveda describes physiology through three doshas – Vata, Pitta and Kapha. A person's birth constitution (prakriti) is a characteristic mix of the three, and the current imbalance (vikruti) guides treatment. Research has started to give these concepts an objective basis. Prasher *et al.* found differences in gene expression and biochemical markers between extreme constitutional types [12], and Govindaraj *et al.* reported genome-wide differences between prakriti groups [13]. Rotti *et al.* developed and evaluated structured instruments for prakriti assessment [6], and Dey and Pahwa reviewed associations between prakriti, metabolism and chronic disease [7]. For this component, the important point is that dosha can be assessed from observable traits such as sleep, appetite, temperature sensitivity and temperament – the same traits a short questionnaire can capture.

2.2.3 Recommender systems

Recommender systems are commonly grouped into collaborative filtering, content-based and knowledge-based approaches, with hybrids combining them [16], [17], [18]. Collaborative filtering needs a history of user interactions, which does not exist for a new platform – the cold-start problem. Knowledge-based recommenders instead match explicit user requirements to item attributes using domain knowledge, which suits rare, high-value decisions such as choosing a treatment centre [17]. Explanations increase users' trust and help them judge recommendations [19]; offline evaluation can then compare variants of a recommender using held-out data or simulated queries [20].

2.2.4 Machine learning and text classification

Random Forests combine many decision trees trained on bootstrap samples with random feature selection, which makes them robust on small, mixed categorical datasets and gives a measure of feature importance [21]. Support Vector Machines (SVM) find a maximum-margin separating hyperplane and work well on sparse, high-dimensional text features [22]. Text is commonly represented with TF-IDF weighting [23]; for languages written without spaces, such as Chinese and Japanese, character n-grams are an effective alternative to word tokens [24]. On small datasets, k-fold and leave-one-out cross-validation give a more reliable estimate of accuracy than a single split [25], although estimates can still be optimistic when model selection uses the same data [26]. Imbalanced classes – such as few Kapha profiles or few unsafe reviews – lower recall on the minority class and need careful metrics [27]. All models in this work use the scikit-learn library [28].

2.3 Related research

Tourism recommender systems. Borràs, Moreno and Valls surveyed intelligent tourism recommenders and found that most rank points of interest, accommodation or routes from user preferences, demographics and context [10]. Health status and constitution are not among the inputs they report. These systems also rely heavily on ratings or interaction histories, which are thin for niche wellness providers.

Health-tourism recommenders. Panteli *et al.* proposed a recommender for health tourism that ranks providers by their closeness to an ideal profile and then selects a diverse final set using a facility-location model [11]. The work shows that health-tourism recommendation benefits from multi-criteria ranking, but it does not use the traveller's constitution, review text or safety signals, and it does not address a specific destination's provider base.

Computational dosha and prakriti assessment. Bidve, Mishra and Annapurna classified prakriti into seven classes, including overlapping dosha types, using chi-square feature selection, K-modes clustering and Multinomial Naïve Bayes, and reported an accuracy of 0.90 on a dataset with 147 questionnaire features [14]. This confirms that dosha can be predicted from questionnaire data, but the long

questionnaire is impractical for a tourist, and the output is a label rather than a recommendation.

Online reviews as quality signals. Hu, Pavlou and Zhang showed that average ratings can misrepresent true quality because of reporting bias, and that the content and distribution of reviews carry additional information [9]. This supports using review text, rather than star ratings alone, to estimate centre quality and to detect safety concerns.

2.4 Existing systems products and techniques

Visitors currently choose wellness centres with general-purpose tools. Table 2.1 compares them with the proposed component on the capabilities that matter for this decision.

Table 2.1: Comparison of existing products with the proposed wellness component

Product / technique	What it offers	What it lacks for this problem
Google Maps and review sites	Location, star rating, photos and free-text reviews for most centres	No condition or constitution matching; quality and safety must be read review by review; no cost estimate
BookRetreats.com [29]	Bookable Ayurveda and meditation retreats in Sri Lanka with descriptions and prices	Covers only listed (mostly premium) retreats; no dosha profiling; no safety screening; filtering is by keyword and date
Online dosha quizzes, e.g. Banyan Botanicals [8]	Self-assessment of dosha through a questionnaire	Result is not linked to any provider, condition or budget; not designed for Sri Lanka
Centre websites and tour operators	Detailed programme information from one provider	No comparison across providers; information quality varies; often English only
SLTDA and official sources	Registration information and statistics	No personalised recommendation
Proposed wellness component	Dosha prediction, condition/budget matching over 83 real centres, review-derived quality and safety, cost and duration estimates, 5 languages, score explanation	No live booking or availability (out of scope)

Techniques found in the literature can be compared in the same way. Knowledge-based, multi-criteria scoring is well suited to a cold-start, high-stakes domain [17],

[11]; Random Forests and SVMs are proven choices for small tabular and text datasets respectively [21], [22]; and per-factor score explanations support trust [19]. The proposed component combines these established techniques in a new configuration for a domain where they have not been applied together.

2.5 Gap synthesis and conceptual framework

Table 2.2 summarises the related work against the six capabilities required to solve the problem in Section 1.3. No reviewed work covers more than two of them.

Table 2.2: Research gap synthesis (✓ = addressed, ~ = partly, – = not addressed)

Work	Health / condition	Dosha profiling	Real SL providers	Review quality / safety	Multilingual	Explainable
Borràs et al. [10]	–	–	–	~	–	~
Panteli et al. [11]	✓	–	–	–	–	~
Bidve et al. [14]	~	✓	–	–	–	–
Fonseka & Pavithra [3]	~	–	~	–	–	–
Hu et al. [9]	–	–	–	✓	–	–
BookRetreats [29]	~	–	✓	~	~	–
This component	✓	✓	✓	✓	✓	✓

The conceptual framework of the component follows directly from this synthesis. Tourist inputs (condition, lifestyle traits, budget, language and optional companion) are transformed by three ML layers – dosha classification, free-text classification and review safety screening – into a profile. A knowledge-based matching model scores every centre on four factors (condition, dosha, quality and budget) and explains each score. The output is enriched with expert knowledge on treatment duration, likely outcome and cost. The research questions in Section 1.6 test each link of this framework, and Chapter 4 turns it into a concrete design.

3 RESEARCH METHODOLOGY

This chapter explains how the research was carried out. It describes the research design, the datasets, the experimental procedure for each research question, the evaluation metrics and the analysis, followed by the ethical, privacy and responsible-AI considerations and the steps taken to make the results reproducible.

3.1 Research design

The component was developed with a **design science research** approach: an artefact (the matching system) is built to solve a practical problem, and the artefact is then evaluated against the research questions. Development was iterative. Each iteration added or improved one capability, ran the automated tests and the model evaluations, and was reviewed with the supervisor at the progress presentations.

The evaluation is quantitative and uses three complementary methods, each mapped to research questions in Table 3.1.

Table 3.1: Research questions, evaluation methods and evidence

RQ	Evaluation method	Evidence produced
RQ1	Six-algorithm comparison on a stratified 80/20 split; 5-fold cross-validation of the selected model; confusion matrix and feature importance	Table 6.1, Figure 6.1–Figure 6.3
RQ2	Offline ablation study: 60 reproducible simulated tourist queries, five model configurations, three outcome metrics	Table 6.2, Figure 6.4
RQ3	Leave-one-out cross-validation of ten per-language classifiers	Table 6.3, Figure 6.5
RQ4	Leave-one-out evaluation of the review classifier against keyword rules on 71 hand-labelled real reviews	Table 6.4
Product	Automated functional, validation, security and admin tests; latency benchmark; user acceptance test protocol	Table 5.3, Table 6.5, Section 6.3

Offline evaluation was chosen because the system is new and has no interaction history, which makes online A/B testing impossible at this stage [20]. Simulated queries let every configuration be compared on identical inputs.

3.2 Data participants or study materials

Five datasets were prepared. They are stored in the repository under `backend/data/` and are summarised in Table 3.2.

Table 3.2: Datasets used by the wellness component

Dataset	Size	Source	Used for
Centre database (`full_candidate_pool.json`)	83 centres; 256 guest reviews	Google Places listings and public reviews, manually checked and annotated	Matching, quality, safety, map, admin
Tourist profiles (`tourist_profiles_training_data.xlsx`)	110 profiles (6 real, 104 expert-rule variations)	Domain expert's patient cases and dosha rules	Dosha classifier (RQ1)
Labelled reviews (`labeled_reviews.json`)	71 reviews (13 concerning, 58 not)	Real reviews from the centre database, labelled by the author	Safety classifier (RQ4)
Symptom and retreat-goal texts (10 files)	109–111 symptom texts and 54 retreat texts per language	Written per language from expert condition descriptions	Text classifiers (RQ3)
Outcome cases (`outcome_training_data.json`)	54 cases (6 real, 48 expert variations)	Domain expert's documented patient outcomes	Outcome predictor

3.2.1 Centre database

The 83 centres comprise 37 curative Ayurveda centres, 32 wellness or rejuvenative centres and 14 spiritual or meditation centres, spread over 46 towns and districts (the largest groups being Negombo with 10, Galle with 8 and Ella with 7). For each centre the database stores the conditions treated, a dosha focus where the centre states one, a price tier (Budget, Budget-Mid, Mid, Mid-Premium or Premium, plus government and donation-based variants), contact details, coordinates, a verification note, the Google rating and review count, and up to four real guest reviews. Thirty-two centres carry a specific verification note – for example a government institution, a named qualified physician or an established meditation tradition (one note records concerns about staff qualifications). The remaining 51 are labelled *Needs verification*, and the interface displays this label.

Each centre has a quality score on a 0–5 scale. For 64 centres with enough review text it was derived from the reviews (labelled *NLP (real reviews)*); for the other 19 the Google rating was used as a fallback, and the interface states which source applies. Scores range from 1.75 to 5.00 with a mean of 4.24.

3.2.2 Tourist profile dataset

Each profile records nationality, age, gender, presenting condition, six questionnaire answers (sleep pattern, digestion and appetite, cold or heat sensitivity and emotional tendency), the primary dosha (vikruti), the recommended treatment and the expected duration. Six profiles are real cases supplied by the project's domain expert, an Ayurvedic practitioner. Because six cases cannot train a classifier, 104 further profiles were generated from the expert's rules: each variation keeps the dominant dosha's characteristic answers for about 70% of traits and draws the remaining 30% from an overlapping dosha, which reflects the mixed constitutions seen in practice. The resulting class distribution is Vata 53, Pitta 33, Kapha 16 and Balanced 8 (Figure 3.1). The eight Balanced profiles were excluded from training because they have no dominant dosha to learn.

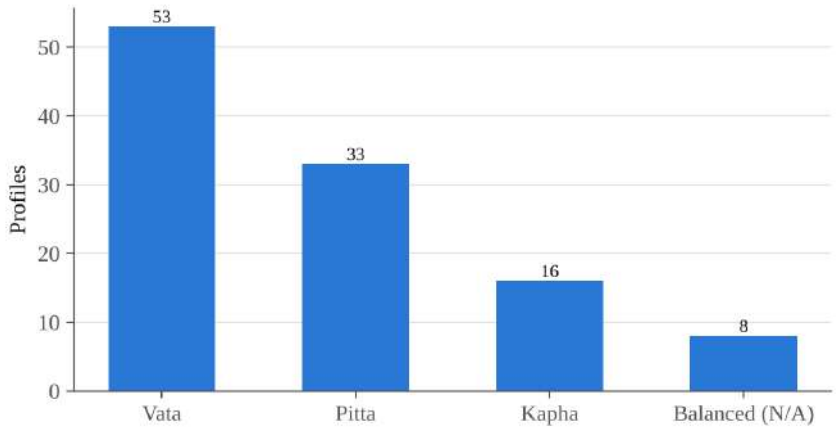


Figure 3.1: Distribution of primary dosha in the 110 tourist profiles

The use of expert-rule variations is an important threat to validity and is discussed in Section 7.5: a model trained on rule-generated data partly learns the rules themselves.

3.2.3 Text and outcome datasets

The symptom datasets contain short first-person descriptions such as *"my knees and fingers ache every morning"* labelled with one of the 12 conditions, prepared separately for each of the five languages so that each model learns its own language's vocabulary (Appendix B describes the AI assistance used). The retreat-goal datasets do the same for six retreat types. The outcome dataset records age, dosha, condition, treatment duration and percentage improvement for six real cases (arthritis, paralysis, sciatica, liver disease, stress and hypertension) and 48 variations of them.

No human participants were recruited for data collection. The real patient cases were provided by the domain expert in anonymised form and contain no names or identifiers (Section 3.6).

3.3 Experimental or study procedure

All models are trained by one notebook, `scripts/Component4_Model_Training.ipynb`, which runs top to bottom in Google Colab or locally, prints every model's parameters and saves the exact files the application loads. The procedure for each research question is as follows.

RQ1 – dosha classifier. (1) Load the 110 profiles and remove the 8 Balanced profiles, leaving 102. (2) Encode the five categorical features with label encoding; age stays numeric. (3) Split 80/20 with stratification, giving 81 training and 21 test profiles with the same class ratio. (4) Train six algorithms with default settings and a fixed random seed (42): Random Forest, Decision Tree, a multilayer perceptron, Gaussian Naïve Bayes, SVM and k-nearest neighbours. (5) Select the best model on test accuracy and weighted F1, and confirm the choice with stratified 5-fold cross-validation on all 102 profiles.

RQ2 – ablation study. (1) Generate 60 tourist queries with a fixed seed, each drawing a condition (from 12), a dosha (from 3) and a budget tier (from 3). (2) For each query, score all 69 Ayurveda-path centres under five configurations: FULL (all four factors), NO-DOSHA, NO-NLP-QUALITY, NO-BUDGET and CONDITION-ONLY. (3) Record the top-ranked centre for each query and configuration and

compute three metrics (Section 3.4). The study is reproduced by `python live_ablation_demo.py --save`.

RQ3 – text classifiers. For each language and domain, fit a TF-IDF vectoriser and a linear SVM, and estimate accuracy with leave-one-out cross-validation; then fit the final model on all examples and save it.

RQ4 – safety classifier. Fit a TF-IDF vectoriser and a class-balanced linear SVM on the 71 labelled reviews, evaluate it with leave-one-out cross-validation, and compare its behaviour with the keyword rules on known difficult cases.

Product evaluation. Run the automated pytest suite and the latency benchmark after every change, and carry out user acceptance testing with representative tourists (Section 6.3).

3.4 Measures and evaluation metrics

Table 3.3: Evaluation metrics

Metric	Definition	Used for
Accuracy	Correct predictions divided by all predictions	RQ1, RQ3, RQ4
Precision, recall, F1 (weighted and per class)	Precision = TP/(TP+FP); recall = TP/(TP+FN); F1 is their harmonic mean. Weighted averages weight each class by its support	RQ1, RQ4
k-fold / LOO CV accuracy	Mean accuracy over held-out folds; LOO holds out one example at a time	RQ1, RQ3, RQ4
Mean absolute error (MAE), R^2	Average absolute difference between predicted and actual improvement (percentage points); proportion of variance explained	Outcome predictor
Average quality of top match	Mean review-derived quality score (0–5) of the top-ranked centre over all queries	RQ2
Budget alignment	Share of queries whose top centre's price tier contains the requested budget tier	RQ2
Dosha compatibility	Share of queries whose top centre's dosha focus equals the query's dosha	RQ2
Median and 95th-percentile latency	Response time of each API endpoint over 200 calls	NFR1
Test pass rate	Passed automated tests divided by executed tests	Product
System Usability Scale (SUS)	Standard 10-item usability questionnaire scored 0–100	UAT (Section 6.3)

Weighted F1 is reported next to accuracy because the dosha classes are imbalanced, and per-class recall is reported because a model can reach high accuracy while missing the minority class [27]. For the safety classifier, recall on the *concerning* class matters more than accuracy, since only 18% of the reviews are concerning.

3.5 Data analysis

All analysis was carried out in Python with scikit-learn 1.8.0, pandas and matplotlib. Classification results are reported as point estimates with the 5-fold standard deviation where available. Given the small test set (21 profiles), one misclassification changes accuracy by about 4.8 percentage points, so differences between algorithms are interpreted together with cross-validation results rather than from the single split alone [25], [26]. Ablation metrics are compared descriptively across configurations on identical queries; each metric difference therefore reflects the configuration, not different inputs. Feature importance is the mean decrease in impurity of the Random Forest [21].

3.6 Ethics privacy security and responsible AI

Health-related data. A tourist's condition and dosha answers are health-related personal data. Sri Lanka's Personal Data Protection Act No. 9 of 2022 (PDPA) treats health data as a special category that needs a lawful basis and appropriate safeguards [30]. The standalone component therefore does not ask for a name, email or any identifier, and it does not store tourist queries; profile answers are used only to compute the response. The integrated platform's `wellness_matching_sessions` table stores sessions only against the platform's tourist identifier (Section 5.3), and storing them requires the platform's consent flow.

Anonymised expert data. The six real patient cases were supplied by the domain expert without names or other identifiers, with only age, condition, dosha, duration and outcome. Guest reviews are publicly available text and are shown with their source; no reviewer names are stored.

Responsible AI. The system is decision support, not diagnosis. Every model output is shown with its uncertainty: the dosha prediction displays its confidence, the free-

text classifiers display their own leave-one-out accuracy and state that the user should confirm the suggestion, the outcome estimate states that it is based on six real cases plus expert variations, and the text classifier returns *no suggestion* rather than guessing when the input does not match the selected language. Matches are explained factor by factor so that the tourist can see why a centre ranked highly [19]. Centres that have not been verified are labelled as such.

Fairness and bias. The dosha model performs worse on the minority Kapha class (Section 6.1), and the text classifiers are weaker in German and Russian than in English, Chinese and Japanese. These differences are reported openly rather than hidden behind an average.

Security. Secrets are kept in environment variables, admin functions require an authenticated session, and inputs are validated (Section 4.5 and Section 5.4).

3.7 Reproducibility and research artefacts

All artefacts are version-controlled in the team's GitHub repository (`Codew-Me/tour_ceylon`, branch `wellness`, folder `components/wellness`). Table 3.4 lists how each reported result can be reproduced.

Table 3.4: Research artefacts and reproduction commands

Artefact	Location	Reproduce with
Training notebook (all 13 models)	`backend/scripts/Component4_Model_Training.ipynb`	Run all cells (Colab or Jupyter)
Trained models	`backend/*.pkl`	Produced by the notebook
Model comparison	`backend/data/model_comparison_results.csv`	Notebook, section 1.3
Training metrics	`backend/data/training_metrics.json`	Notebook, section 6
Ablation results	`backend/data/ablation_results.json`	`python live_ablation_demo.py --save`
Automated tests	`backend/tests/`	`python -m pytest tests -v`
Latency benchmark	`backend/tests/perf_benchmark.py`	`python tests/perf_benchmark.py`

Every random process uses the fixed seed 42, and scikit-learn is pinned to version 1.8.0 so that saved models load identically on any machine. Re-running the notebook reproduces all reported metrics exactly; the latency benchmark depends on the machine and is reported with its environment.

4 PRODUCT AND SOLUTION DESIGN

This chapter presents the design of the wellness component. It lists the functional and non-functional requirements, shows the component's architecture and its external interfaces within the complete Tour Ceylon solution, records the main design decisions, and describes the detailed design of the matching model, the data model and the user interface.

4.1 Requirements

Requirements were derived from the problem statement, the literature review, the domain expert's input and the needs of the integrated platform. Table 4.1 lists the functional requirements (FR) and Table 4.2 the non-functional requirements (NFR). Each requirement is traced to the tests that verify it.

Table 4.1: Functional requirements

ID	Requirement	Verified by
FR1	Provide separate Ayurveda and spiritual paths with their own conditions and centres	TC01–TC05
FR2	Predict the tourist's dominant dosha and confidence from age, gender and four lifestyle answers	TC06, TC08, TC39
FR3	Rank centres by condition, dosha, quality and budget, return the top results and explain each score	TC06–TC13
FR4	Support a travelling companion with a second condition and a combined score	TC14
FR5	Compare two scenarios (different condition or budget) and show full centre details	TC15–TC19
FR6	Estimate treatment duration and trip cost for a number of travellers, in several currencies	TC20–TC22
FR7	Suggest a condition or retreat type from free text in five languages, with a no-signal response for unmatched text	TC23–TC27
FR8	Flag safety-relevant reviews and predict a personalised treatment outcome	TC28–TC32
FR9	Provide an authenticated admin dashboard with analytics, centre CRUD and flagged-review moderation	TC40, TC44–TC48
FR10	Show results on a map and offer practical guidance (before-you-go checklist, seasonal advice, sharing)	UI walkthrough (Section 5.4)

Table 4.2: Non-functional requirements (quality characteristics follow ISO/IEC 25010 [31])

ID	Quality characteristic	Requirement and target	Verified by
NFR1	Performance efficiency	Recommendation response in under 200 ms (median) on a laptop-class machine	Benchmark (Table 6.5)
NFR2	Reliability	Invalid input returns HTTP 400 with a message, never a server error; unseen answer values do not crash prediction	TC33–TC39
NFR3	Security	Admin API requires a session; secrets in environment variables; HttpOnly session cookie; source and .env files not served	TC40–TC43
NFR4	Usability	Three-step wizard, five interface languages, voice input, results understandable without Ayurvedic knowledge	UAT (Section 6.3)
NFR5	Transparency	Each match shows its factor breakdown; model outputs show confidence or accuracy	TC12, TC27, TC31
NFR6	Portability	Runs with PostgreSQL or, if the database is unavailable, from a JSON file; runs on Windows and Linux	Section 5.5
NFR7	Maintainability	Modular code; automated tests; models retrainable from one notebook	Section 3.7
NFR8	Reproducibility	Fixed seeds and pinned library versions so results can be regenerated exactly	Section 3.7

4.2 Component architecture within the complete solution

Figure 4.1 shows the architecture of the wellness component. It follows a three-tier design: a browser-based presentation layer, a REST API layer and a data layer, with the matching engine and the ML models between the API and the data.

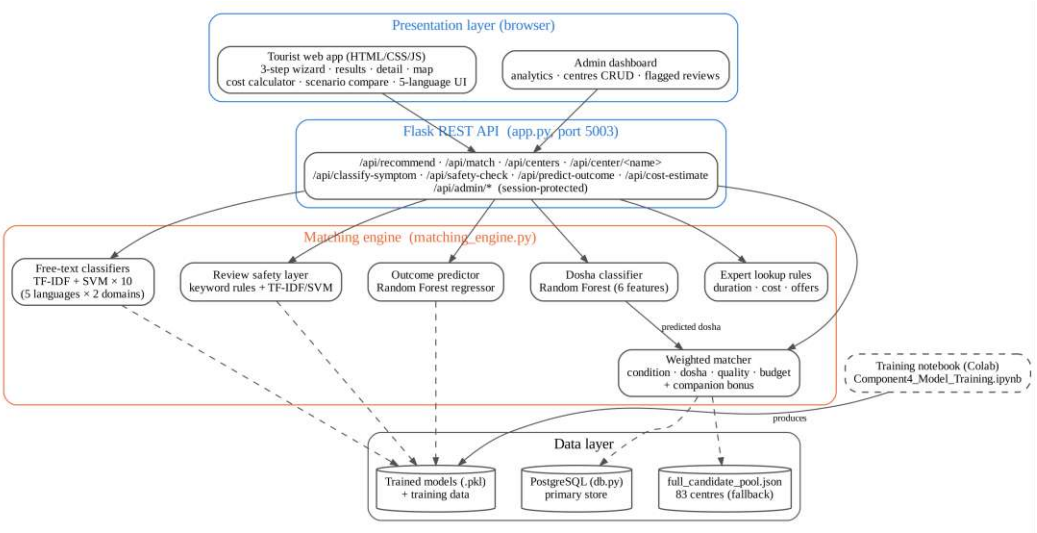


Figure 4.1: Architecture of the wellness component

The **presentation layer** contains the tourist web application and the admin dashboard. Both are static HTML, CSS and JavaScript files served by the same Flask process, so no cross-origin configuration is needed. The **API layer** (`app.py`) exposes JSON endpoints, validates input and enforces admin authentication. The **matching engine** (`matching_engine.py`) holds the domain logic: the dosha classifier, the weighted matcher, the text and safety classifiers, the outcome predictor and the expert lookup rules for duration, cost and offers. The **data layer** (`db.py`) loads centres from PostgreSQL and falls back automatically to the JSON file if the database is not configured or not reachable. Trained models are stored as files and loaded lazily on first use.

4.2.1 External interfaces

The component communicates with four external parties, summarised in Table 4.3.

Table 4.3: External interfaces of the wellness component

Interface	Direction and protocol	Purpose
Tour Ceylon shared backend	Flask blueprint registered under `/api` in the unified application; same JSON contract as the standalone API	Lets the platform's home page and other components call wellness endpoints
Shared PostgreSQL database	SQL via psycopg2; migration `005_wellness.sql`	Stores centres, reviews, condition mappings and matching sessions linked to the platform's tourist identifier

Interface	Direction and protocol	Purpose
OpenStreetMap tiles via Leaflet 1.9.4	HTTPS tile requests from the browser	Map view of matched centres
Browser Web Speech API	In-browser speech recognition	Voice input of symptoms in the five languages

Table 4.4 documents the REST interface. All endpoints accept and return JSON and follow resource-oriented REST conventions [32].

Table 4.4: REST API of the wellness component

Method and endpoint	Input	Output
GET /api/health	–	Status and number of centres loaded
GET /api/conditions?path=	path: ayurveda or spiritual	Condition or retreat types for the path
GET /api/centers?path=	path	Centre names, categories and districts
POST /api/recommend	condition, age, gender, four lifestyle answers, budget_tier, optional path, top_n, companion_condition	Predicted dosha and confidence, duration, outcome, ranked centres with score breakdown and risk flags
POST /api/match	condition, dosha, budget_tier, optional path, top_n	Ranked centres for a known dosha (scenario comparison)
GET /api/center/<name>	Centre name	Full centre detail, risk flags, offers or session pattern
POST /api/cost-estimate	condition, price_tier, travelers	Minimum and maximum days and cost
POST /api/classify-symptom	text, lang (en, de, ru, zh, ja), domain	Suggested category, confidence, model accuracy, no-signal flag
POST /api/safety-check	reviews (list of text)	Flags from keyword rules and from the ML model, with confidence
POST /api/predict-outcome	age, dosha, condition, duration_days	Predicted improvement (%) and model accuracy
/api/admin/*	Session cookie after POST /api/admin/login	Login, logout, analytics, centre CRUD, flagged reviews

4.3 Design alternatives and decisions

Table 4.5 records the main design decisions, the alternatives considered and the reason for each choice.

Table 4.5: Design decisions and justification

Decision	Alternatives considered	Choice and justification
Recommendation approach	Collaborative filtering; content-based similarity; learning-to-rank	Knowledge-based weighted scoring. There is no interaction history (cold start), the decision is rare and high-value, and every score must be explainable [17], [19]
Dosha classifier	Decision Tree, neural network, Naïve Bayes, SVM, k-NN	Random Forest: best test accuracy (0.810) and F1 (0.794) of six algorithms, robust on 102 mixed categorical records, and gives feature importance [21]
Text classifiers	Pre-trained multilingual transformer; keyword lists; one shared model with translation	TF-IDF + linear SVM per language: works with about 100 examples, runs in under 2 ms on a CPU, needs no translation service; character n-grams for Chinese and Japanese [22], [24]
Review safety	Keyword rules only; ML only	Hybrid: rules give high recall on known terms, the ML model catches idiomatic false positives such as "cannot recommend it highly enough"
Matching weights	Equal weights; weights learned from data	Expert-set weights (condition 0.35, quality 0.25, dosha 0.20, budget 0.20), validated by the ablation study; no labelled relevance data exists to learn them
Spiritual path weights	Same four weights as Ayurveda	Dosha removed and its weight redistributed (condition 0.45, quality 0.30, budget 0.25) because all 14 spiritual centres share the same dosha focus, so the dosha term cannot discriminate
Database	MongoDB; JSON only; SQLite	PostgreSQL to match the team's shared database, with an automatic JSON fallback for demos and grading
Front end	React single-page app	Plain HTML, CSS and JavaScript served by Flask: no build step, same origin as the API; the platform shell links to it

4.4 Detailed design

4.4.1 Matching pipeline

Figure 4.2 shows the processing pipeline of one recommendation request.

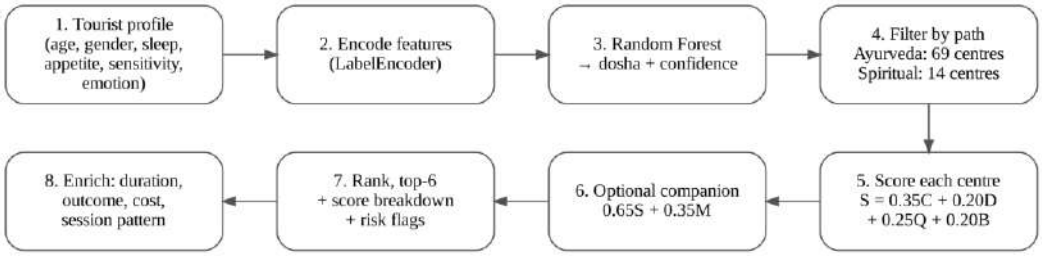


Figure 4.2: Processing pipeline of a recommendation request

For the Ayurveda path, the score of centre *c* for a tourist with condition *k*, predicted dosha *d* and budget tier *b* is:

$$S(c) = 0.35 \cdot C(c,k) + 0.20 \cdot D(c,d) + 0.25 \cdot Q(c) + 0.20 \cdot B(c,b) \quad (4.1)$$

where C = 1.0 if the centre lists the condition and 0.3 otherwise; D = 1.0 if the centre's dosha focus equals d, 0.6 if the centre has a general focus and 0.4 otherwise; Q is the quality score divided by 5; and B = 1.0 if the price tier contains the requested tier and 0.5 otherwise. The partial values (0.3, 0.6, 0.4, 0.5) keep near-matches in the ranking instead of excluding them. For the spiritual path the dosha term is removed: $S = 0.45 \cdot C + 0.30 \cdot Q + 0.25 \cdot B$. When a companion's condition m is given, the combined score is:

$$S_combined(c) = 0.65 \cdot S(c) + 0.35 \cdot M(c,m) \quad (4.2)$$

where M is 1.0 if the centre also treats the companion's condition and 0.3 otherwise. Each term is returned separately as a percentage contribution, so the interface can show, for example, that a centre earned 35 of a possible 35 points for condition and 19 of 25 for quality.

4.4.2 Data model

The PostgreSQL schema has four tables: `centers` (one row per centre with all attributes), `reviews` (review text linked to a centre, with a risk flag), `condition_dosha_mapping` (expert mapping of conditions to the dosha usually involved and typical treatments) and `matching_sessions` (one row per search, linked to the platform's `tourist_id`). In the shared platform database the same tables are created with a `wellness_` prefix by migration `005_wellness.sql` so that they do not clash with other components.

4.4.3 User interface design

The tourist interface is a guided three-step wizard (Figure 4.3). The tourist first chooses a path, then (1) states the condition, age and gender – optionally by typing or speaking a description, which the text classifier turns into a suggested condition – (2) answers four lifestyle questions in plain language ("How do you usually sleep?") instead of Ayurvedic terms, and (3) chooses a budget. Figure 4.4 shows the first two steps. The language selector switches all labels between English, German, Russian, Chinese and Japanese.

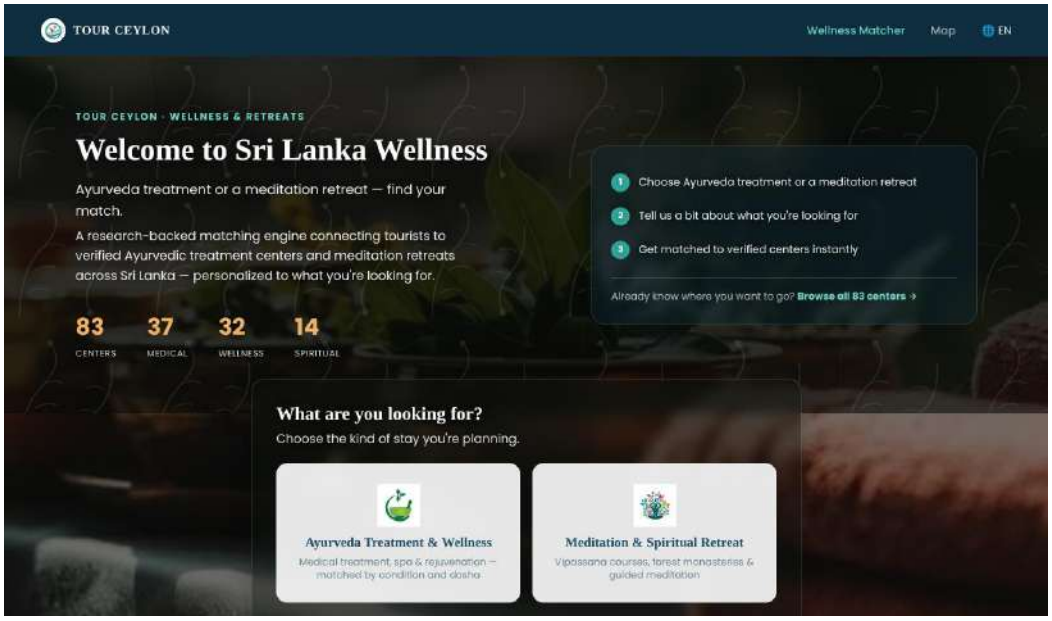


Figure 4.3: Landing page with the choice between the Ayurveda and spiritual paths

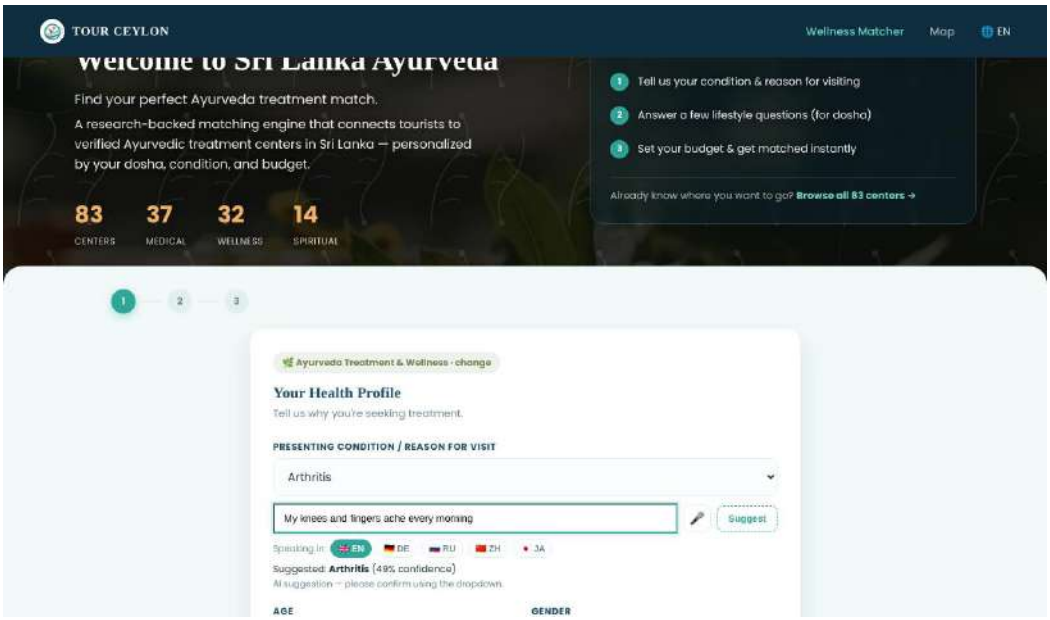


Figure 4.4: Wizard step 1 with a free-text symptom suggestion ("Arthritis, 49% confidence")

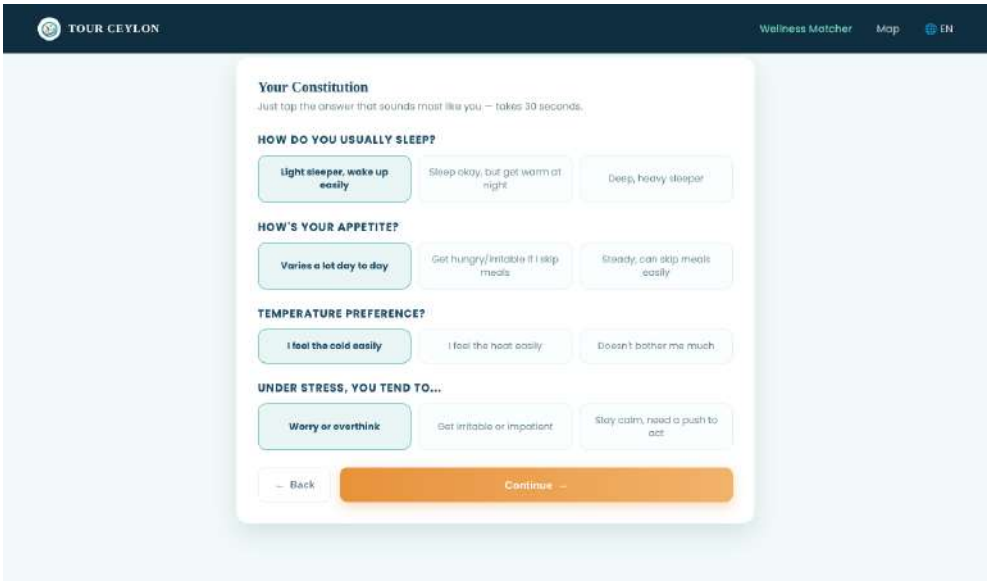


Figure 4.5: Wizard step 2: plain-language lifestyle questions used to predict the dosha

4.5 Standards and professional practice

- **Software quality.** Non-functional requirements are organised by the ISO/IEC 25010 quality characteristics [31].
- **Web security.** The design addresses the relevant OWASP Top 10 (2021) risks [33]: broken access control (A01) through session checks on every admin endpoint; cryptographic failures and security misconfiguration (A02, A05)

through secrets in environment variables, a random session key and HttpOnly, SameSite cookies; injection (A03) through parameterised SQL values and a fixed whitelist of updatable columns in `db.py`; and vulnerable components (A06) through pinned library versions.

- **API design.** Resource-oriented REST endpoints with JSON bodies and standard HTTP status codes (200, 400, 401, 404) [32].
- **Data protection.** Data minimisation and purpose limitation in line with the PDPA [30] (Section 3.6).
- **Version control and review.** Git feature branches per component, descriptive commits and pull requests into the shared repository.
- **Testing.** Automated regression tests run before every commit; each test case carries an identifier traced to a requirement (Table 4.1).
- **Responsible AI.** Uncertainty is shown with every prediction, and the system states that it does not replace a physician (Section 3.6).

5 IMPLEMENTATION INTEGRATION AND TESTING

This chapter describes how the design was implemented, integrated and verified. It covers the technology stack, the implementation of each part of the component, the integration with the shared platform, the automated tests and the deployment of the component.

5.1 Development environment and technology stack

Table 5.1 lists the technologies used and the reason each was chosen. Library versions are pinned in `requirements.txt`.

Table 5.1: Technology stack and justification

Layer	Technology (version)	Justification
Language	Python 3.11–3.13	Standard language for ML; same language for API and models avoids a separate model server
Web framework	Flask 3.1	Lightweight, matches the shared Tour Ceylon backend, so the component can be registered as a blueprint
Machine learning	scikit-learn 1.8.0 [28]	Mature implementations of Random Forest, SVM and TF-IDF; small CPU models; version pinned so saved models load identically
Data handling	pandas 3.0, NumPy 2.5, openpyxl 3.1	Tabular processing and reading the Excel training file
Database	PostgreSQL with psycopg2; JSON fallback	Relational store shared with the platform; JSON fallback for offline demos
Configuration	python-dotenv	Keeps database credentials and admin secrets out of the code
Front end	HTML5, CSS3, JavaScript (ES2020)	No build tooling; served by Flask on the same origin
Maps	Leaflet 1.9.4 with OpenStreetMap tiles	Free and key-less, unlike commercial map APIs
Voice input	Web Speech API	Built into Chromium browsers; supports all five languages
Model training	Jupyter notebook on Google Colab	Reproducible, shareable, runs without local setup
Testing	pytest 9 with the Flask test client; Playwright (browser check)	Fast in-process API tests; scripted browser walkthrough

Layer	Technology (version)	Justification
Version control	Git and GitHub	Branch per component, history as evidence of work

Development was carried out on Windows 11 with Visual Studio Code and PowerShell. Tests and benchmarks in this thesis were run on a Linux machine with two Intel Xeon 2.8 GHz virtual cores and Python 3.13.

5.2 Component implementation

The backend consists of 1,246 lines of Python across four modules, and the front end of 2,833 lines of HTML, CSS and JavaScript (Table 5.2).

Table 5.2: Size of the implementation

File	Lines	Responsibility
`backend/app.py`	352	REST API, input validation, admin authentication, static file serving
`backend/matching_engine.py`	619	Dosha model, matching, text and safety classifiers, outcome predictor, duration and cost rules, analytics
`backend/db.py`	137	PostgreSQL access with automatic JSON fallback
`backend/live_ablation_demo.py`	138	Reproducible ablation study
`backend/tests/` (4 files)	386	Automated tests and latency benchmark
`frontend/js/app.js`	1,845	Wizard, results, details, map, cost calculator, scenario comparison, translations, voice input
`frontend/index.html`, `css/style.css`	598	Page structure and styling
`frontend/admin/` (3 files)	390	Admin dashboard

5.2.1 Dosha classifier

The dosha model is trained when the server starts, from the Excel profile file, with the same features and parameters as in the training notebook. This guarantees that the deployed model and the evaluated model are identical. Unseen answer values are mapped to a known category instead of raising an error (TC39). The prediction returns the dosha and the model's class probability as a confidence percentage.

```
DOSHA_MODEL = RandomForestClassifier(n_estimators=100, random_state=42)
DOSHA_MODEL.fit(_X, _y_enc)

def predict_dosha(age, gender, sleep, appetite, sensitivity, emotion):
    row = pd.DataFrame([[age, gender, sleep, appetite, sensitivity,
emotion]],
                       columns=FEATURE_COLS)
    ...  # encode each answer; unseen values fall back to a known class
    proba = DOSHA_MODEL.predict_proba(row)[0]
    return dosha, round(float(max(proba)) * 100, 1)
```

5.2.2 Matching engine

`match_centers()` implements Equations 4.1 and 4.2. It filters the centre pool by path, computes the four factor values, stores each factor's contribution with the result, attaches the keyword risk flags and returns the top *n* centres sorted by score (or by combined score when a companion is given). The results are then enriched with the expected duration, the outcome likelihood and the cost estimate.

5.2.3 Multilingual text classifiers

Ten classifiers (five languages × two domains) are loaded lazily and cached. A key implementation detail is the *no-signal* check: if the input text shares no real vocabulary with the selected language's model, the endpoint returns no suggestion instead of the model's arbitrary default class. For Chinese and Japanese, which use character n-grams, the check ignores the whitespace n-gram that scikit-learn's `char_wb` analyser adds at word boundaries, because it would otherwise match any text.

```
X = vectorizer.transform([text])
names = vectorizer.get_feature_names_out()
real_matches = [i for i in X.nonzero()[1] if names[i].strip() != ""]
if not real_matches:
    return {"predicted_condition": None, "no_signal": True, ...}
```

5.2.4 Review safety, outcome and cost

The safety layer runs both the keyword rules and the SVM on each review and reports which method flagged it, so the admin can see where they disagree. The outcome predictor is a Random Forest regressor saved together with its encoders and its own leave-one-out MAE and R^2, so the API reports the model's accuracy from the model file rather than from text written in the code. The cost estimate multiplies the expert duration range for the condition by a daily rate for the price tier and the

number of travellers; the interface converts it from USD into EUR, GBP and RUB using fixed illustrative exchange rates.

5.2.5 Web interface and admin dashboard

The interface implements the wizard described in Section 4.4.3, a results view with factor bars for every match (Figure 5.1), a detail view, a map view, a cost calculator, a scenario comparison, a browse view of all 83 centres, a before-you-go checklist, seasonal guidance for the west/south and east/north coasts, and sharing by WhatsApp or e-mail. All labels are translated into the five languages. The admin dashboard (Figure 5.2) shows live analytics, the model comparison and ablation results read from the repository files, centre CRUD and the flagged-review list.

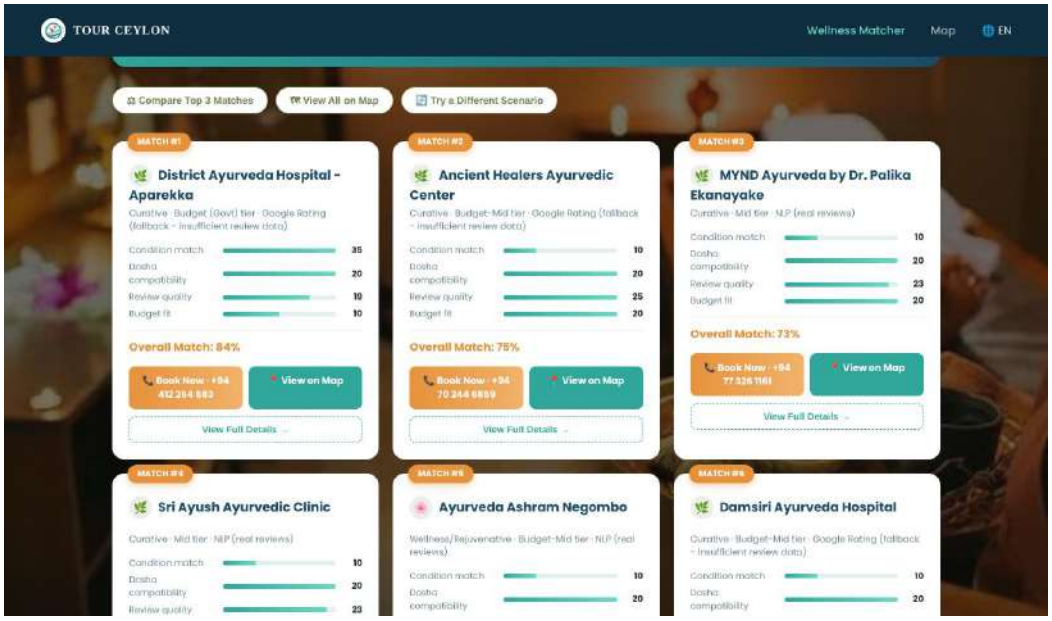


Figure 5.1: Ranked results with the per-factor score breakdown for each centre

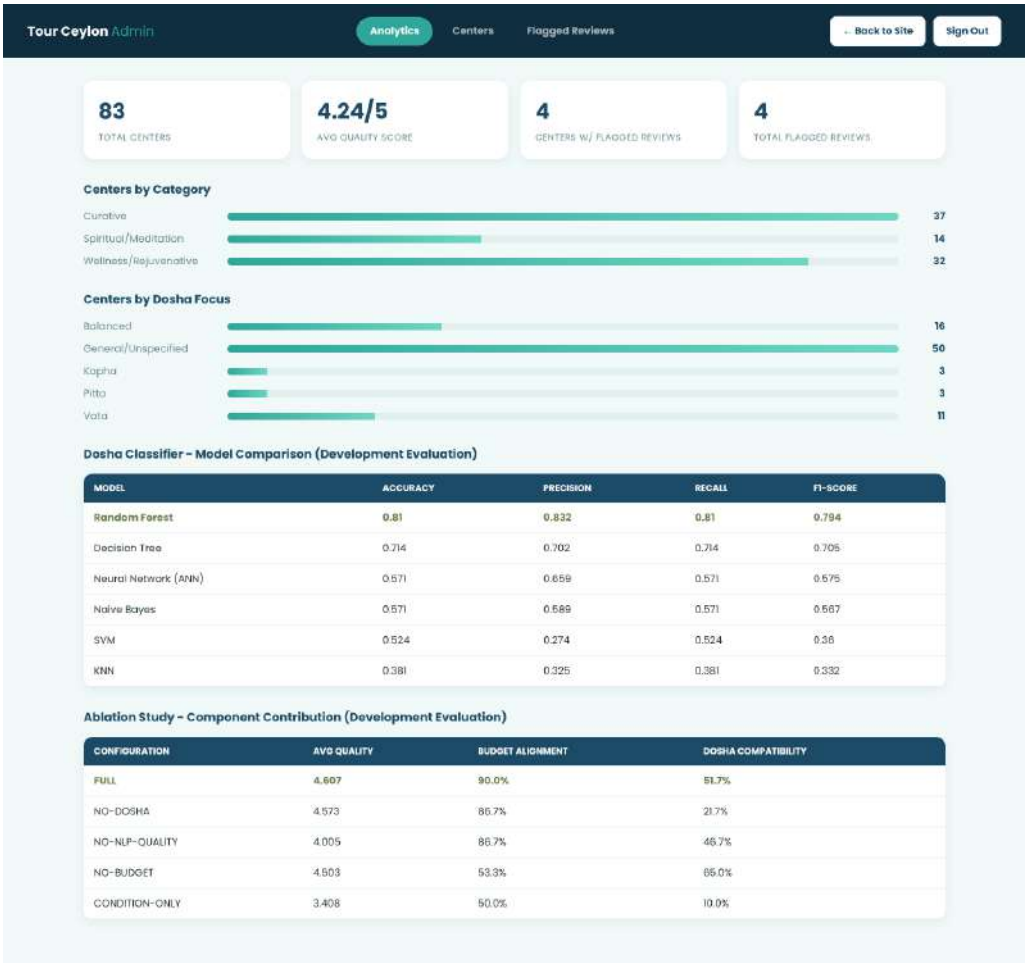


Figure 5.2: Admin dashboard: analytics, model comparison and ablation study

5.2.6 Improvements made after Progress Presentation 2

Feedback at the second progress presentation questioned whether reported model metrics were computed or hard-coded. In response, every model was retrained from raw data in one notebook that prints each model's parameters and computes all metrics live; the outcome model's accuracy is now stored inside the model file; the ablation script writes the results that the admin dashboard displays; and secrets were moved from the code to environment variables. Re-running the notebook reproduces every metric in this thesis exactly (Section 3.7).

5.3 Integration with the other components

The wellness component is integrated into the Tour Ceylon platform at three levels.

1. **Backend.** The unified Tour Ceylon Flask application registers the wellness routes as a blueprint under `/api`, using the same JSON contract as the standalone

API. Other components and the platform's health check (`components: travel, sos, wellness, risk`) therefore reach the wellness functions through one backend.

2. **Database.** Migration `005_wellness.sql` in the shared backend creates `wellness_centers`, `wellness_reviews`, `wellness_condition_mappings` and `wellness_matching_sessions` in the shared PostgreSQL database. `wellness_matching_sessions` links each search to the platform's `tourist_id`, which is the integration point with the itinerary component: a matched centre and treatment duration can be used when the tourist's itinerary is planned.
3. **Front end.** The platform's home page shows a Wellness tile that opens the wellness interface, configured through `SITES.wellness` in the shared front-end configuration, and the admin password is configured through the shared variable `WELLNESS_ADMIN_PASSWORD`.

While preparing this thesis an integration risk was identified: the shared backend pins `scikit-learn==1.5.1`, whereas the retrained wellness models are saved with version 1.8.0. Models saved with one scikit-learn version may fail to load in another (for example, a support vector model raised `AttributeError: 'SVC' object has no attribute '_effective_probability'` in a mismatched environment). The fix – pinning 1.8.0 in the shared `requirements.txt` – was communicated to the integration owner (Section 6.4).

5.4 Verification and software testing

Verification used four complementary activities: (1) automated tests, (2) model validation (Chapter 6), (3) a performance benchmark and (4) a scripted browser walkthrough.

5.4.1 Automated tests

The test suite uses pytest with the Flask test client, so every test exercises the real routing, validation, matching and models without a running server. Tests that change data (admin create, edit and delete) back up the centre file and restore it afterwards, so tests never alter the real dataset. Table 5.3 lists all 48 test cases. Parameterised

cases (for example TC09, which runs once for each of the 12 conditions) expand to **75 executions**, and **all 75 pass** (`75 passed in 0.82 s`).

Table 5.3: Automated test cases (all passed)

ID	Req.	Scenario	Expected result
TC01	FR1	Health endpoint reports the dataset	200; 83 centres loaded
TC02	FR1	Tourist interface is served	200
TC03	FR1	Ayurveda conditions listed	12 conditions incl. Arthritis
TC04	FR1	Spiritual retreat types listed	6 types
TC05	FR1	Path separation of centres	69 Ayurveda and 14 spiritual centres; no overlap
TC06	FR2, FR3	Recommendation for a standard profile	Valid dosha, confidence in (0, 100], scores sorted descending
TC07	FR3	Default number of results	6 recommendations
TC08	FR2	Same input twice	Identical dosha and ranking (deterministic)
TC09 (×12)	FR3	Every Ayurveda condition	6 results and a duration range for each
TC10 (×6)	FR1, FR3	Every spiritual retreat type	Only spiritual centres; dosha not scored
TC11 (×3)	FR3	Budget, Mid and Premium requests (Stress)	All top-6 centres in the requested tier
TC12	FR3, NFR5	Score breakdown	Four factor contributions sum to the match score (±2 rounding)
TC13	FR3	Condition match ranks first	Top centre treats arthritis
TC14	FR4	Companion with Stress	Companion match 1.0 or 0.3; combined score = $0.65 \cdot S + 0.35 \cdot M$
TC15	FR5	Match with a known dosha	200; one result
TC16	FR5	Two different scenarios	Different centre lists
TC17	FR5	Ayurveda centre detail	Typical offers and risk flags returned
TC18	FR5	Spiritual centre detail	Session pattern returned
TC19	FR5	Unknown centre	404
TC20	FR6	Cost for 1 and 2 travellers	Cost doubles; minimum ≤ maximum
TC21	FR6	Premium versus Budget cost	Premium is more expensive
TC22	FR6	Cost for unknown condition	404

ID	Req.	Scenario	Expected result
TC23 (×5)	FR7	Native symptom text in EN, DE, RU, ZH, JA	A valid Ayurveda condition for each language
TC24	FR7	Japanese text with English model	No-signal response, no guess
TC25	FR7	Retreat goal (spiritual domain)	A valid retreat type
TC26	FR7, NFR2	Empty text	400
TC27	FR7, NFR5	Disclaimer content	Reports leave-one-out accuracy and example count
TC28	FR8	Clearly concerning review	Flagged by keyword rules
TC29	FR8	"Cannot recommend it highly enough"	Not flagged by the ML model
TC30	FR8	Empty review list	Empty flag list
TC31	FR8, NFR5	Outcome prediction	Personalised; 0–100%; MAE read from the model file
TC32	FR8	Outcome for a condition outside the training cases	Falls back to the static estimate
TC33 (×4)	NFR2	Age "abc", 0, 150, missing	400 with message
TC34	NFR2	Missing required fields	400 listing the fields
TC35	NFR2	Invalid path (recommend and match)	400
TC36 (×3)	NFR2	top_n = 0, 500, "six"	400
TC37	NFR2	Match with missing fields	400
TC38	NFR2	Cost with missing fields	400
TC39	FR2, NFR2	Unseen answer value	200 (handled, no crash)
TC40	NFR3	Admin endpoints without login	401 for list, analytics, flags, create, delete
TC41	NFR3	Wrong admin password	401
TC42	NFR3	Path traversal to .env and source files	Not served
TC43	NFR3	Session cookie flags	HttpOnly set
TC44	FR9	Admin list and analytics	83 centres; model comparison and ablation present
TC45	FR9	Add, edit and delete a centre	Centre appears, is edited, then is removed

ID	Req.	Scenario	Expected result
TC46	FR9, NFR2	Add a centre without a name	400
TC47	FR9	Flagged-review moderation list	At least one flagged centre
TC48	FR9, NFR3	Logout	Admin endpoints return 401 again

Table 5.4 summarises the coverage by category.

Table 5.4: Test coverage by category

Category	Test cases	Executions	Passed
Functional – recommendation and paths (FR1–FR5)	19	37	37
Functional – cost, text, safety, outcome (FR6–FR8)	13	17	17
Input validation and robustness (NFR2)	7	12	12
Security (NFR3)	4	4	4
Admin workflow (FR9)	5	5	5
Total	**48**	**75**	**75**

5.4.2 Browser walkthrough

A scripted Playwright run opened the interface in Chromium, switched the language to German, completed the wizard and requested a match. The run produced a dosha result and a ranked list with no error dialogs, which confirms that the front end and the validated API work together in a non-English interface. The same run captured the screenshots used in this thesis.

5.4.3 Defects found and fixed

Table 5.5: Defects found during verification

Defect	Found by	Fix
Non-numeric age caused an HTTP 500 server error	Negative test TC33	Age validated (whole number, 1–120) with an HTTP 400 message
Admin password and session key written in the code	Security review	Moved to environment variables; random session key if none is set; HttpOnly, SameSite cookie

Defect	Found by	Fix
Admin dashboard highlighted the worst ablation configuration because rows were sorted alphabetically	Visual check of the dashboard	Rows shown in a fixed order with FULL first
Ablation figures in the dashboard differed from a fresh run	Re-running the ablation script	Script now writes the file the dashboard reads (`--save`)
Models failed to load when scikit-learn was not pinned	Fresh install	`scikit-learn==1.8.0` pinned in requirements

5.5 Deployment and operation

The component runs as a single Flask process that serves both the API and the interface on port 5003:

```
cd components/wellness/backend
pip install -r requirements.txt
python app.py            # http://localhost:5003  (admin: /admin)
```

Configuration is read from `backend/.env` (template: `.env.example`): PostgreSQL connection settings, `WELLNESS_ADMIN_PASSWORD` and `FLASK_SECRET_KEY`. If PostgreSQL is not configured or reachable, the server logs a warning and loads the 83 centres from the JSON file, so the system can always be demonstrated. Admin edits are written to whichever source is active.

Operationally, models are retrained by running the notebook and copying the produced files into `backend/`; the admin dashboard then shows the new comparison results without code changes. Within the platform, the shared backend is started once and serves all components. For production, the shared backend already includes Gunicorn as a WSGI server; a containerised deployment with managed PostgreSQL and HTTPS termination is planned as future work (Section 9.4).

6 RESULTS AND VALIDATION

This chapter reports the results of the evaluation. It answers the four research questions, presents the product and performance results, describes the user and stakeholder validation and the integration results, and summarises the findings.

6.1 Research results

6.1.1 RQ1 – dosha classification

Figure 6.1 compares the six algorithms on the same 21-profile test set. Random Forest achieved the highest accuracy (0.810), precision (0.832), recall (0.810) and F1 score (0.794), followed by the Decision Tree (accuracy 0.714). The neural network and Naïve Bayes reached 0.571, SVM 0.524 and k-nearest neighbours 0.381.

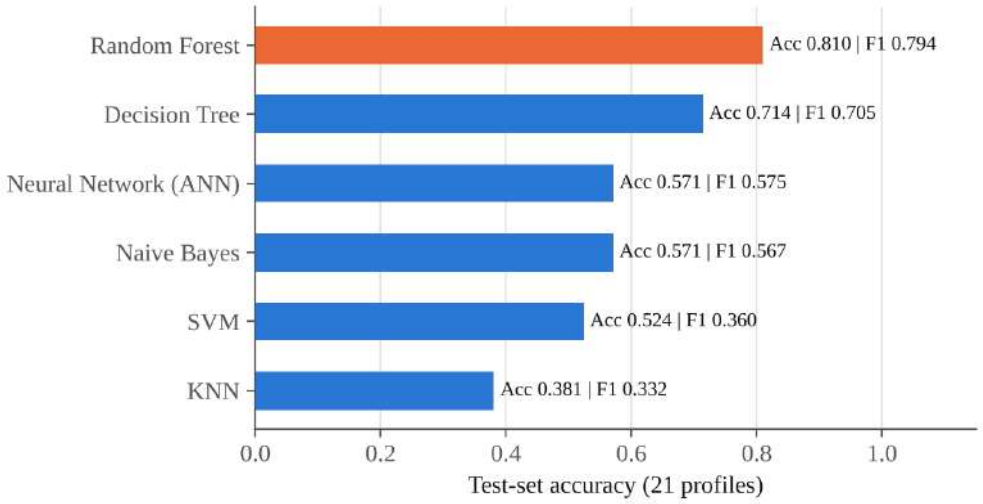


Figure 6.1: Test accuracy and weighted F1 of six algorithms for dosha classification

Stratified 5-fold cross-validation of the Random Forest on all 102 profiles gave fold accuracies of 100.0%, 71.4%, 90.0%, 80.0% and 80.0%, a mean of **84.3% ± 9.8%**. The cross-validated mean is close to the test-set accuracy, which indicates that the test result is not an accident of one split. Table 6.1 shows the per-class results and Figure 6.2 the confusion matrix.

Table 6.1: Per-class results of the Random Forest dosha classifier (test set, n = 21)

Class	Precision	Recall	F1	Support
Kapha	1.00	0.33	0.50	3

Class	Precision	Recall	F1	Support
Pitta	0.86	0.86	0.86	7
Vata	0.77	0.91	0.83	11
Macro average	0.88	0.70	0.73	21
Weighted average	0.83	0.81	0.79	21



Figure 6.2: Confusion matrix of the Random Forest dosha classifier (test set)

Vata and Pitta are recognised well. Kapha, the smallest class (16 profiles, 3 in the test set), has perfect precision but a recall of only 0.33: two of three Kapha tourists were predicted as Vata. Figure 6.3 shows that sleep pattern (0.235), cold or heat sensitivity (0.229) and emotional tendency (0.213) are the most informative features, followed by digestion and appetite (0.173) and age (0.129); gender contributes little (0.022).

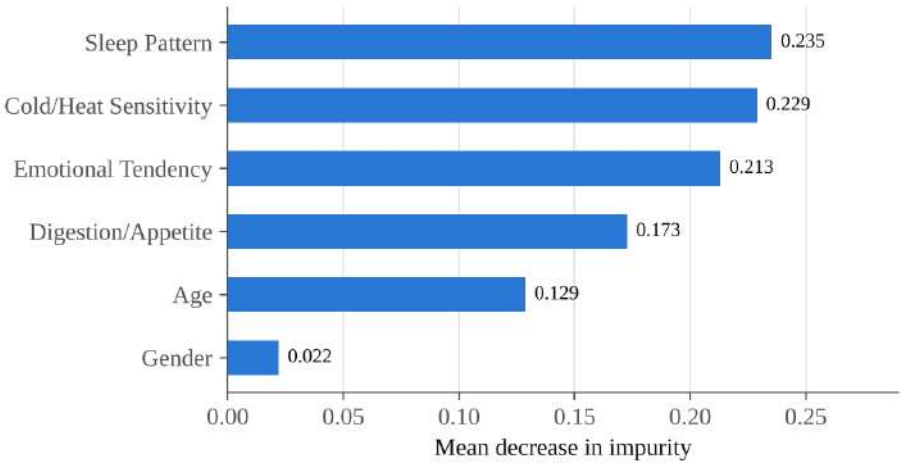


Figure 6.3: Random Forest feature importance for dosha prediction

6.1.2 RQ2 – contribution of the matching factors (ablation study)

Table 6.2 and Figure 6.4 report the ablation study over 60 simulated queries on the 69 Ayurveda-path centres.

Table 6.2: Ablation study of the matching model (60 queries, seed 42)

Configuration	Avg. quality of top match (/5)	Budget alignment	Dosha compatibility
FULL (condition + dosha + quality + budget)	**4.607**	**90.0%**	51.7%
NO-DOSHA	4.573	86.7%	21.7%
NO-NLP-QUALITY	4.005	86.7%	46.7%
NO-BUDGET	4.503	53.3%	65.0%
CONDITION-ONLY	3.408	50.0%	10.0%

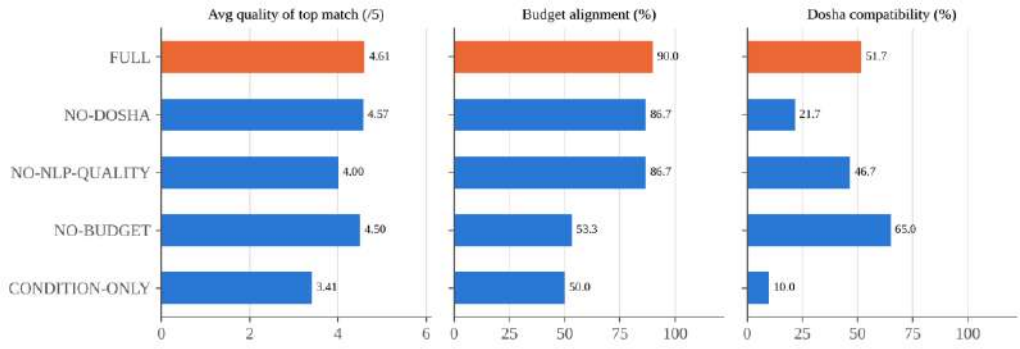


Figure 6.4: Ablation study: effect of removing each factor (FULL highlighted)

The full model outperforms condition-only matching on all three metrics: the top match's average quality rises from 3.41 to 4.61, budget alignment from 50.0% to 90.0% and dosha compatibility from 10.0% to 51.7%. Hypothesis H1 is therefore supported. Each factor is responsible for the metric it targets: removing quality lowers average quality by 0.60 points, removing dosha cuts dosha compatibility from 51.7% to 21.7%, and removing budget cuts budget alignment from 90.0% to 53.3%. Removing budget raises dosha compatibility to 65.0%, which shows a trade-off between the two factors (Section 7.1).

6.1.3 RQ3 – multilingual text classifiers

Table 6.3: Leave-one-out accuracy of the ten text classifiers

Language	Analyser	Symptom: examples	Symptom: LOO accuracy (12 classes)	Retreat: examples	Retreat: LOO accuracy (6 classes)
English	word 1–2-gram	109	66.1%	54	79.6%
German	word 1–2-gram	109	54.1%	54	38.9%
Russian	word 1–2-gram	111	60.4%	54	42.6%
Chinese	character 1–2-gram	109	73.4%	54	87.0%
Japanese	character 1–2-gram	109	66.1%	54	75.9%

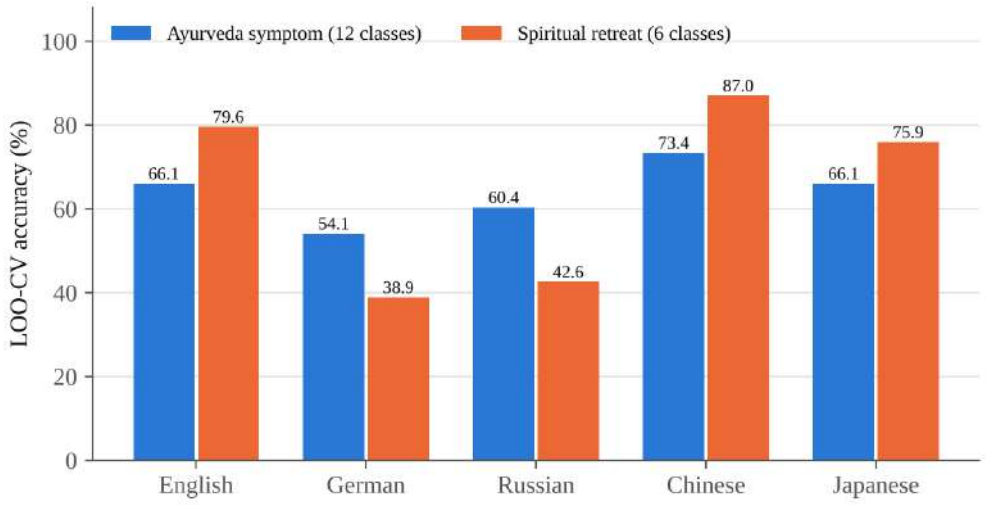


Figure 6.5: Leave-one-out accuracy of the symptom and retreat-goal classifiers by language

All ten classifiers perform well above chance (8.3% for 12 symptom classes and 16.7% for 6 retreat types). Chinese is the strongest language in both domains (73.4% and 87.0%), English and Japanese are close behind, and German and Russian are the weakest, particularly for retreat goals (38.9% and 42.6%).

6.1.4 RQ4 – review safety screening

Table 6.4: Leave-one-out evaluation of the review safety classifier (71 real reviews)

Metric	Value
Accuracy	0.831
Precision (concerning)	1.000
Recall (concerning)	0.077 (1 of 13)
F1 (concerning)	0.143
Confusion matrix [TN FP; FN TP]	[58 0; 12 1]

The classifier never raises a false alarm (precision 1.0) but finds only one of the 13 concerning reviews in leave-one-out evaluation. Its value is complementary: it correctly classifies the positive idiom *"I cannot recommend it highly enough"*, which the keyword rules flag because it contains *"not recommend"* (TC29). In the live database the keyword rules flag 4 reviews at 4 centres, which the admin dashboard lists for manual review. The hybrid design therefore uses the rules as the primary detector and the model as a second opinion.

6.1.5 Treatment outcome predictor

The Random Forest regressor predicts treatment improvement with a leave-one-out **MAE of 4.7 percentage points** and **R^2 = 0.624** on 54 cases (Figure 6.6). The six real cases lie close to the diagonal. The model is used only for the six conditions it was trained on; for other conditions the system falls back to an expert estimate and labels it as such (TC32).

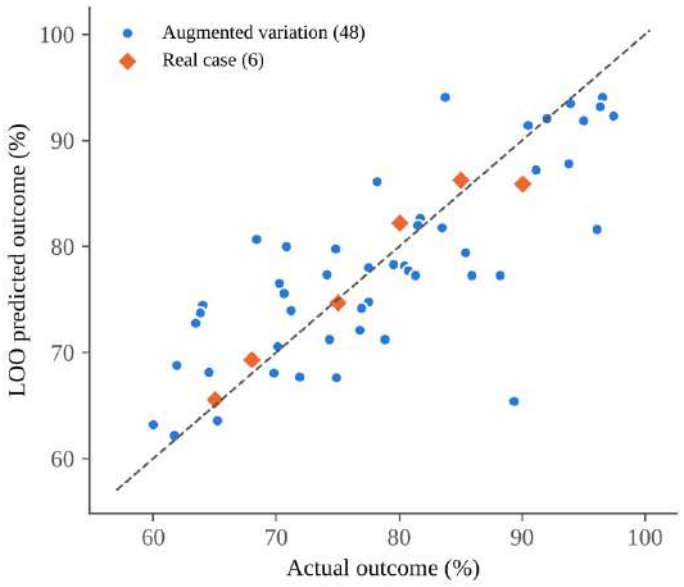


Figure 6.6: Outcome predictor: actual versus leave-one-out predicted improvement

6.2 Product and technical results

All ten functional requirements were implemented. Table 6.5 shows the latency of each endpoint measured with `tests/perf_benchmark.py` (200 calls per endpoint after 5 warm-up calls).

Table 6.5: API latency (in-process, 2 vCPU Linux, Python 3.13)

Endpoint	Median (ms)	95th percentile (ms)
POST /api/recommend	22.19	26.94
POST /api/match	1.78	2.16
POST /api/classify-symptom	1.56	1.94
POST /api/safety-check	1.72	2.34
POST /api/predict-outcome	10.30	16.02
POST /api/cost-estimate	0.41	0.60

The full recommendation, including dosha prediction, scoring of all centres and enrichment, takes 22 ms at the median and 27 ms at the 95th percentile, well within the 200 ms target (NFR1). Server start-up, which includes training the dosha model, takes about 2 seconds. Network time is not included because the benchmark runs in-process; in a local deployment it adds only a few milliseconds.

The automated suite passes 75 of 75 executions (Section 5.4), and the scripted browser walkthrough completed without errors in German. Table 6.6 summarises the status of the requirements.

Table 6.6: Requirement status

Requirement	Status	Evidence
FR1–FR9	Met	TC01–TC48 pass
FR10 (map, checklist, seasonal advice, sharing)	Met	Browser walkthrough; screenshots
NFR1 Performance	Met	Median 22 ms (target < 200 ms)
NFR2 Reliability	Met	TC33–TC39
NFR3 Security	Met (demo-grade auth)	TC40–TC43; single shared admin password
NFR4 Usability	Pending user validation	Section 6.3
NFR5–NFR8	Met	Sections 3.7, 5.2

6.3 User or stakeholder validation

Evaluation (Sections 6.1 and 6.2) measures how well the system performs; validation asks whether it solves the real problem for real users. Validation is organised in two parts.

Domain expert validation. The domain expert provided the six real patient cases, the condition–dosha–treatment mapping and the duration and cost structure on which the expert rules are based. [TO BE COMPLETED: if the domain expert reviews the final system, record the date, what was reviewed and the feedback here.]

User acceptance testing (UAT). UAT uses realistic tasks rather than opinion questions. Each participant (target: at least 8 foreign or foreign-resident visitors interested in Ayurveda or meditation, plus 2 centre operators) completes five tasks without help, after which they complete the System Usability Scale (SUS) questionnaire:

1. Find a centre that treats a given condition within a given budget, and explain why it was ranked first.

2. Change the interface to your own language and describe your symptoms in free text.
3. Compare two scenarios (another condition or budget) and choose one.
4. Estimate the cost of a three-week programme for two people in your currency.
5. Find a Vipassana retreat and state when courses usually start.

Table 6.7: UAT results

Task	Completion rate	Mean time	Observations
T1 Condition and budget match	[__ / __]	[__ s]	[to be completed after UAT]
T2 Language and free text	[__ / __]	[__ s]	[to be completed after UAT]
T3 Scenario comparison	[__ / __]	[__ s]	[to be completed after UAT]
T4 Cost estimate	[__ / __]	[__ s]	[to be completed after UAT]
T5 Spiritual retreat	[__ / __]	[__ s]	[to be completed after UAT]
SUS score (0–100)	[mean ± SD]	–	[n = __ participants]

[TO BE COMPLETED BEFORE SUBMISSION: replace the bracketed values with the real UAT results, add one paragraph interpreting them, and attach the consent form and questionnaire in Appendix A. If UAT cannot be completed, state this as a limitation in Section 7.5 instead.]

6.4 Integration results

Integration was verified at the interface level. The standalone component passes all 75 tests through the same JSON contract that the platform blueprint exposes, the shared database migration creates the four wellness tables, and the platform home page links to the wellness interface.

The version mismatch described in Section 5.3 (scikit-learn 1.5.1 in the shared backend versus 1.8.0 for the wellness models) is the one open integration issue. [TO BE COMPLETED: after the shared `requirements.txt` is updated, run the wellness tests against the unified backend and record the result here, for example "all wellness endpoints returned HTTP 200 in the unified application on <date>".]

6.5 Summary of findings

Table 6.8: Summary of findings

Question / objective	Finding
RQ1 / O2	Random Forest is the best of six algorithms: accuracy 0.810, F1 0.794, 5-fold CV 84.3% ± 9.8%. Sleep, temperature sensitivity and temperament are the most informative traits; Kapha recall is low (0.33).
RQ2 / O3	The full four-factor model improves average match quality (3.41 → 4.61), budget alignment (50.0% → 90.0%) and dosha compatibility (10.0% → 51.7%) over condition-only matching. H1 is supported.
RQ3 / O4	Per-language TF-IDF + SVM classifiers reach 54–73% (12 classes) and 39–87% (6 classes) LOO accuracy; Chinese is strongest, German and Russian weakest.
RQ4 / O5	Keyword rules detect known terms; the ML model has precision 1.0 but recall 0.077, and is useful only as a second opinion that removes idiomatic false positives.
O5	Outcome predictor: LOO MAE 4.7 points, R^2 0.624 on 54 cases.
O1, O6	83 real centres; full-stack system with 48 test cases (75/75 pass), median recommendation latency 22 ms, integrated into the platform backend and database.

7 DISCUSSION

This chapter interprets the results, compares them with the literature and existing solutions, states the research and product contributions, and discusses the limitations, threats to validity and wider implications of the work.

7.1 Interpretation of findings

Dosha classification (RQ1). Random Forest outperformed simpler models because the relationship between answers and dosha is a combination of several traits rather than one decisive answer, which an ensemble of trees captures well on a small, categorical dataset [21]. Distance-based k-NN and the RBF-kernel SVM performed poorly because label-encoded categories have no meaningful distances. The feature ranking is consistent with how Ayurveda describes the doshas: sleep quality, temperature sensitivity and temperament are the classic distinguishing signs of Vata, Pitta and Kapha, while gender carries little information [6]. The weak Kapha recall (0.33) is a direct result of class imbalance – Kapha has 16 profiles against 53 for Vata – and the classifier favours the majority class when traits overlap [27]. Because the interface displays the predicted dosha with its confidence and allows the tourist to try other scenarios, a misclassification lowers the dosha term of the score (at most 20 of 100 points) but does not remove suitable centres from the list.

Matching factors (RQ2). The ablation study shows that each factor does the job it was designed for and that the factors combine without cancelling each other out. The large quality gain over condition-only matching (3.41 to 4.61 out of 5) matters most for the tourist's safety and satisfaction, because condition matching alone often surfaces poorly reviewed centres that happen to list the condition. The rise in dosha compatibility when budget is removed (51.7% to 65.0%) reveals a real trade-off: centres with a specific dosha focus are often not in the requested price tier, so honouring the budget sometimes means accepting a general-focus centre. With equal 0.20 weights the system resolves this trade-off in favour of budget, which matches the practical reality that a tourist cannot use an unaffordable centre. A related observation from testing is that condition dominates budget for rare conditions: for

arthritis, a government hospital in the Budget tier ranks first even for Mid and Premium requests, because few centres list arthritis. This is the intended behaviour – a suitable centre is shown before an affordable but unsuitable one – but it is explained to the user through the score breakdown.

Multilingual classifiers (RQ3). Character n-grams suit Chinese and Japanese, where a short phrase already contains several informative characters, and these languages achieved the highest accuracies. German and Russian are highly inflected – the same word appears in many forms – so word n-grams built from about 100 sentences fragment into rare features; compound nouns in German add to this. The weak retreat-goal results in these two languages are therefore a data and representation problem rather than an implementation fault. The no-signal check is as important as accuracy: it stops the system from returning a confident-looking guess when the tourist writes in a language other than the one selected (TC24).

Safety screening (RQ4). With only 13 concerning reviews, the classifier learns too little about the many ways a review can express a concern, and it errs on the side of not flagging. This makes it unsuitable as a primary detector, but useful as a precise second opinion. The finding supports the hybrid design rather than an ML-only design.

7.2 Comparison with literature baselines and existing solutions

Bidve *et al.* reported an accuracy of 0.90 for prakriti classification with Multinomial Naïve Bayes on a questionnaire of 147 features [14]. The dosha classifier in this work reaches 0.81 (cross-validated 0.84) with only six features. The figures are not directly comparable – the datasets, classes and labels differ – but the comparison shows the design trade-off: a much shorter questionnaire that a tourist can complete in about 30 seconds, at a moderate cost in accuracy. In this work Naïve Bayes performed poorly (0.571), which suggests that the strong independence assumption suits a long questionnaire of many weak indicators better than a short one of a few strongly related traits.

Panteli *et al.* rank health-tourism providers by their closeness to an ideal provider and then diversify the result [11]. The wellness component similarly uses multi-

criteria ranking, but adds personalisation by constitution, review-derived quality and safety, and a transparent per-factor explanation, which the explanation literature identifies as important for trust [19]. Compared with the tourism recommenders surveyed by Borràs *et al.* [10], the component needs no interaction history, which solves the cold-start problem for a new platform.

Compared with existing products (Table 2.1), the component is the only option that connects a dosha assessment to real providers, and the only one that summarises quality and safety across reviews. It does not provide live booking, which BookRetreats does, and it covers 83 centres rather than every centre in the country.

7.3 Research contribution

- A **constitution-aware matching model** for wellness tourism that combines condition, dosha, review-derived quality and budget, with an ablation study quantifying each factor's contribution – a combination not found in the reviewed literature.
- Empirical evidence that a **six-feature questionnaire** is sufficient for useful dosha prediction (0.81 test accuracy, 84.3% cross-validated), with sleep, temperature sensitivity and temperament as the most informative traits.
- A comparison of **per-language lightweight text classifiers** for five tourist languages, showing that character n-grams perform best for Chinese and Japanese and identifying inflected languages as the weak point.
- Evidence that a **hybrid keyword and ML approach** is needed for review safety screening when labelled data is scarce.
- A **curated, annotated dataset** of 83 real Sri Lankan Ayurveda and meditation centres with verification notes and review-derived quality scores.

7.4 Product contribution

The product contribution is a working, tested, multilingual web application that a tourist can use today to find, compare and cost Ayurveda centres and meditation retreats, and an admin dashboard that lets an operator maintain the centre data and monitor flagged reviews. The component is integrated into the Tour Ceylon platform's backend and database, so the wellness choice can feed into the tourist's

itinerary. Every model can be retrained and every reported metric reproduced from one notebook.

7.5 Limitations and threats to validity

Construct and internal validity.

- **Augmented training data.** Of the 110 dosha profiles, 104 are variations generated from the domain expert's rules, and the same applies to 48 of the 54 outcome cases. A classifier trained on rule-generated data partly learns the rules, so its accuracy on real tourists may be lower than reported. This is the most important limitation of the study, and collecting real assessed profiles is the first item of future work.
- **Vikruti from a questionnaire.** The model predicts the current imbalance from self-reported traits; a practitioner's pulse diagnosis and examination are not captured.
- **Expert-set weights.** The matching weights and partial-match values were set with expert judgement and validated by ablation, not learned from tourist choices.

Statistical conclusion validity. The dosha test set has only 21 profiles (3 Kapha), so a single error changes accuracy by about 4.8 points, and the 5-fold standard deviation is 9.8 points. The algorithm comparison used default hyperparameters and the same data for model selection and evaluation, which can make the estimate optimistic [26]. The ablation queries are simulated, not real searches.

External validity and data. The database covers 83 centres, not every centre in Sri Lanka; 51 still need verification; quality scores for 19 centres fall back to Google ratings; and the script that computed the review-derived quality scores during data preparation is not part of the current repository, so those scores cannot be regenerated from code. Reviews were collected from Google, and commercial use of stored Google review text requires review of the platform's terms (Section 8.4). Cost estimates use simplified daily rates and fixed exchange rates.

Product limitations. Admin authentication uses one shared password rather than individual accounts; UAT with real tourists is [pending / reported in Section 6.3];

latency was measured on one machine; and the free-text classifiers support five languages only.

7.6 Implications

For tourists, the component turns an opaque, high-stakes choice into a guided decision in their own language, with the reasons for each recommendation visible. **For Ayurveda centres**, it rewards good service through review-derived quality and gives smaller, well-reviewed centres visibility next to large brands. **For the sector and policy makers**, the verification labels and safety flags illustrate the kind of structured information that practitioners and researchers have called for [5], [3]. **For research**, the results suggest that constitution-aware recommendation is feasible with short questionnaires, and that the main barrier is the availability of real labelled data rather than the choice of algorithm.

8 COMMERCIALISATION AND SUSTAINABILITY

This chapter examines the commercial potential and sustainability of the wellness component: its value proposition and target users, its market and competitive position, a proposed business model, scalability, intellectual property and compliance, and its contribution to the Sustainable Development Goals.

8.1 Value proposition and target users

Value proposition. *"Find the right Ayurveda centre or meditation retreat in Sri Lanka for your condition, constitution and budget – in your language, with honest quality and safety information and a clear cost estimate."*

Table 8.1: Target user segments and value delivered

Segment	Need	Value delivered
International wellness tourists (primary)	Choose a suitable, safe centre before travelling	Personalised, explained matches; safety flags; cost in own currency; five languages
Ayurveda centres and retreats	Reach qualified foreign guests	Visibility based on service quality; verified-listing badge; guest pre-qualification by condition
Tour operators and travel agents	Package wellness stays reliably	Matching API; verified provider data
Tour Ceylon platform	A differentiating wellness feature	Integrated wellness module linked to itinerary planning

The primary segment is sizeable and valuable: SLTDA recorded 12,977 visitors whose main purpose was health or Ayurveda in 2025, and these visitors stayed 14.7 days on average – longer than any other group – with Germany alone contributing 6,363 of them [1]. The interface languages were chosen to match the main source markets, and German was given particular attention because of its share.

8.2 Market and competitive position

The global wellness tourism market was worth USD 830.2 billion in 2023 and is projected to reach USD 1,351.0 billion by 2028 [2]. In Sri Lanka, wellness tourists rank Ayurveda and holistic treatments as the destination's most important capability

[3], while the supply side remains fragmented and inconsistently regulated [5]. A trusted matching service addresses this gap between strong demand and weak information.

Table 8.2: Competitive position

Competitor	Strength	Weakness addressed by the component
Google Maps and review sites	Universal coverage, free	No personalisation; no condition or dosha matching; reviews unstructured
BookRetreats and similar portals [29]	Online booking, premium inventory	Limited, mostly premium inventory; no constitution profiling or safety screening
Dosha quiz websites [8]	Popular self-assessment	Not linked to providers or to Sri Lanka
Tour operators	Personal service	Limited to partner centres; not transparent; costly

The defensible advantages are the curated and verified Sri Lankan dataset, the constitution-aware matching model and its explanations, the safety layer, and native-language support – all of which take time and domain access to replicate.

8.3 Business or sustainability model

A freemium, two-sided model is proposed. Tourists use the matching service free of charge, which builds traffic and trust. Revenue comes from the supply side and from partners:

- **Booking commission** on enquiries that convert into stays, collected from centres;
- **Verified listing subscription** for centres that pass verification (registration, qualified physician, hygiene check), which adds a badge and richer profile content;
- **B2B API licence** for tour operators and travel platforms that want to embed the matching service.

Table 8.3 gives an illustrative revenue scenario. The market figure comes from SLTDA [1]; the adoption rate, average booking value, commission rate and subscription fee are **assumptions** that must be validated with centres before any business decision.

Table 8.3: Illustrative annual revenue scenario (assumptions marked *)

Item	Conservative	Moderate
Health / Ayurveda visitors per year [1]	12,977	12,977
Share who book through the platform*	5% (649)	10% (1,298)
Average booking value per visitor*	USD 1,000	USD 1,000
Commission rate*	10%	10%
Commission revenue	USD 64,900	USD 129,800
Verified listings × USD 20 per month*	30 centres: USD 7,200	60 centres: USD 14,400
Total annual revenue	**USD 72,100**	**USD 144,200**

Operating costs are low because the models run on a CPU and the application needs a single small server and a managed PostgreSQL instance; the main costs are centre verification visits, data maintenance and marketing in the source markets. **Sustainability** of the service depends on keeping the data fresh, which the admin dashboard supports, and on centres seeing enough qualified enquiries to justify a subscription.

8.4 Scalability intellectual property and compliance

Scalability. The matching engine scores all centres in about 22 ms, and scoring grows linearly with the number of centres, so even a database of several thousand centres would stay within interactive limits. The stateless API can be replicated behind a load balancer, and the same design extends to other Ayurveda destinations such as Kerala by adding a centre dataset and translations. New languages require only a training file per language.

Intellectual property. The source code, the curated dataset with its verification annotations and the matching model are original work. Third-party components are used under permissive licences (scikit-learn, Flask and Leaflet are BSD-licensed), and OpenStreetMap map data requires attribution under the Open Database Licence.

Compliance.

- **Data protection.** Health-related inputs are special-category personal data under the PDPA [30]. A commercial service must obtain explicit consent, publish a

privacy notice, minimise stored data and secure it; the current design already avoids storing identifiers in the standalone component.

- **Medical claims.** The service must keep presenting itself as decision support, show disclaimers, and avoid promising treatment outcomes.
- **Provider regulation.** Verification should confirm each centre's registration with the Department of Ayurveda and the qualifications of its physicians.
- **Third-party data.** Review text and ratings collected from Google are subject to Google's terms of service; a commercial version should use licensed data or collect its own verified guest reviews.

8.5 Contribution to the Sustainable Development Goals

Table 8.4: Contribution to the Sustainable Development Goals [34]

SDG	Contribution
SDG 3 – Good health and well-being	Helps people find appropriate, safe traditional health care; surfaces safety concerns and unverified providers
SDG 8 – Decent work and economic growth (target 8.9: sustainable tourism that creates jobs and promotes local culture and products)	Directs high-value, long-stay tourists to local Ayurveda centres, including small and regional providers, and promotes Sri Lanka's traditional medicine heritage
SDG 12 – Responsible consumption and production (target 12.b: monitor sustainable tourism impacts)	Transparent quality and safety information supports responsible choices; analytics give a view of the wellness tourism supply

9 CONCLUSION AND FUTURE DIRECTIONS

This chapter concludes the thesis. It summarises how the research problem was addressed, evaluates the achievement of each objective, restates the individual contributions and outlines directions for future work.

9.1 Conclusion

This thesis addressed the difficulty international visitors face in choosing a suitable, safe Ayurveda centre or meditation retreat in Sri Lanka. The Wellness Component of the Tour Ceylon platform solves this problem by combining ML-based dosha profiling, a transparent four-factor matching model, review-derived quality and safety signals, multilingual input and output, and duration, outcome and cost estimates, over a curated database of 83 real centres.

The evaluation answered all four research questions. A Random Forest predicts the dominant dosha from six questionnaire features with 0.810 accuracy and 84.3% cross-validated accuracy. The full matching model improves the quality, budget fit and dosha fit of the top recommendation over condition-only matching, supporting hypothesis H1. Lightweight per-language classifiers handle free-text input in five languages, best in Chinese and English, weakest in German and Russian. Review safety needs a hybrid of rules and ML, because the ML model alone has very low recall on the available data. The system meets its functional and performance requirements, passes all 75 automated test executions and is integrated with the platform's backend and database. The main limitation is the reliance on expert-rule variations of six real cases for training, which makes validation with real tourists the most important next step.

9.2 Achievement of objectives

Table 9.1: Achievement of objectives

Objective	Status	Evidence
O1 Curated centre dataset	Achieved	83 centres, 256 reviews, verification notes (Section 3.2.1)

Objective	Status	Evidence
O2 Dosha classifier	Achieved	Six-algorithm comparison; RF accuracy 0.810, CV 84.3% (Section 6.1.1)
O3 Weighted matching and ablation	Achieved	Equations 4.1–4.2; ablation study (Table 6.2)
O4 Multilingual access	Achieved (weaker in DE/RU)	Ten classifiers; five-language interface (Table 6.3)
O5 Safety, duration, outcome and cost	Achieved (safety ML recall low)	Sections 6.1.4–6.1.5; TC20–TC32
O6 Tested, secure, integrated application	Achieved; UAT and unified-backend test pending	75/75 tests; Sections 5.3, 6.3, 6.4

9.3 Individual research and product contributions

The author's individual contributions are the curated centre dataset; the dosha classification study and model; the constitution-aware matching model and its ablation study; ten multilingual text classifiers and the hybrid safety layer; the outcome predictor; the complete Flask and web implementation with its admin dashboard; the reproducible training notebook; the automated test suite and benchmark; and the integration of the component into the Tour Ceylon platform (Section 1.8).

9.4 Future directions

1. **Real-world data.** Collect dosha assessments by qualified practitioners for real tourists (with consent) to retrain and externally validate the classifier, and record actual treatment outcomes for the outcome model.
2. **Learning from use.** Log anonymised selections and enquiries to learn matching weights from real choices, and evaluate them with online A/B tests [20].
3. **Better text models.** Expand the German and Russian training sets, add lemmatisation, and evaluate small multilingual transformer models once more data is available.
4. **Safety data.** Label more reviews, especially concerning ones, to raise the safety classifier's recall; add aspect-level analysis (hygiene, staff, treatment).

5. **Data coverage and freshness.** Complete verification of the remaining 51 centres, add the review-quality pipeline to the repository, and refresh reviews periodically.
6. **Production deployment.** Containerise the service, deploy it with managed PostgreSQL and HTTPS, replace the shared admin password with per-user accounts, and add live exchange rates and booking enquiries.

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GLOSSARY

Ayurveda – the traditional system of medicine of South Asia, practised in Sri Lanka as Deshiya Chikitsa alongside Ayurveda proper and regulated by the Department of Ayurveda.

Dosha – one of the three functional principles of Ayurveda (Vata, Pitta and Kapha) used to describe a person's constitution and current imbalance.

Vata – the dosha associated with movement; typical descriptors include light or disturbed sleep, variable appetite, sensitivity to cold and a tendency to worry.

Pitta – the dosha associated with metabolism and heat; typical descriptors include strong or irregular appetite, sensitivity to heat and irritability.

Kapha – the dosha associated with structure and stability; typical descriptors include deep sleep, steady appetite and a calm temperament.

Prakriti – a person's inherent, birth constitution expressed as a combination of the three doshas.

Vikruti – a person's current state of dosha imbalance; the target variable of the dosha classifier in this work.

Panchakarma – a set of five Ayurvedic cleansing and rejuvenation procedures, usually delivered over several weeks.

Shirodhara – an Ayurvedic therapy in which warm oil is poured in a steady stream over the forehead.

Vipassana – an insight-meditation tradition; in Sri Lanka commonly taught as fixed-length, silent 10-day residential courses.

Ablation study – an evaluation in which components of a system are removed one at a time to measure each component's contribution.

Leave-one-out cross-validation – an evaluation in which each example is held out once and predicted by a model trained on all the other examples.

Data augmentation (expert-rule) – the creation of additional training records by systematically varying real cases according to rules supplied by a domain expert.

APPENDICES

Appendix A Extended results and reproduction guide

A.1 Reproducing the results

```
# 1. Train all models (prints parameters and metrics):
#    open backend/scripts/Component4_Model_Training.ipynb
#    in Google Colab, run all cells, upload training_data.zip,
#    copy the .pkl files to backend/, the .csv to backend/data/
# 2. Re-run the ablation study (writes data/ablation_results.json)
cd components/wellness/backend
python live_ablation_demo.py --save
# 3. Run the automated tests and the latency benchmark
python -m pytest tests -v
python tests/perf_benchmark.py
```

A.2 Example API exchange

Request to `POST /api/recommend` and the response (fields shortened to one recommendation):

```
{"condition": "Arthritis", "age": 52, "gender": "M",
 "sleep_pattern": "Light/disturbed sleep", "appetite": "Variable
appetite",
 "cold_heat_sensitivity": "Sensitive to cold",
 "emotional_tendency": "Anxious under pain", "budget_tier": "Mid",
"top_n": 1}
```

```
{"predicted_dosha": "Vata", "dosha_confidence_pct": 98.0,
 "expected_duration": {"range": "18-25 days",
   "source": "Domain expert case data (21-day real patient case)"},
 "recommendations": [{
   "name": "District Ayurveda Hospital - Aparekka", "district": "Matara",
   "category": "Curative", "price_tier": "Budget (Govt)", "verified":
"Yes (Govt)",
   "match_pct": 84, "condition_pct": 35, "dosha_pct": 20,
   "quality_pct": 19, "budget_pct": 10, "risk_flags": []}]}
```

A.3 Model parameters

Table A.1: Parameters of the trained models (non-default values in bold)

Model	Algorithm	Key parameters
Dosha classifier	RandomForestClassifier	**n_estimators = 100**, criterion = gini, max_features = sqrt, max_depth = None, min_samples_leaf = 1, bootstrap = True, **random_state = 42**
Symptom / retreat classifiers (×10)	TfidfVectorizer + SVC	**ngram_range = (1, 2)**, **max_features = 500**, **analyzer = word (EN/DE/RU) or char_wb (ZH/JA)**, **stop_words = english (EN only)**; SVC **kernel = linear**, C = 1.0, **probability = True**, **random_state = 42**

Model	Algorithm	Key parameters
Safety classifier	TfidfVectorizer + SVC	**ngram_range = (1, 2)**, **stop_words = english**, **max_features = 500**; SVC **kernel = linear**, **class_weight = balanced**, **probability = True**
Outcome predictor	RandomForestRegressor	**n_estimators = 100**, max_depth = None, min_samples_leaf = 1, **random_state = 42**

A.4 Per-fold cross-validation of the dosha classifier

Table A.2: Stratified 5-fold cross-validation (shuffle, seed 42)

Fold	1	2	3	4	5	Mean ± SD
Accuracy	100.0%	71.4%	90.0%	80.0%	80.0%	84.3% ± 9.8%

[TO BE ADDED IF UAT IS CARRIED OUT: participant information sheet, consent form, task sheet and SUS questionnaire.]

Appendix B Academic integrity and AI use disclosure

Generative AI tools were used during this project in accordance with the module policy. The author remains responsible for all content, code, data, analyses and conclusions, and has verified every number in this thesis against the repository and its test outputs.

Table B.1: Use of generative AI tools

Tool	Purpose	How the output was verified	Sections / artefacts affected
Claude (Anthropic)	Assistance with code (API validation, test suite, benchmark, training notebook), debugging and data-handling scripts	All code reviewed by the author, run locally, and checked by the automated tests (75/75) and by re-running the notebook	`backend/`, `tests/`, `scripts/`
Claude (Anthropic)	Drafting and editing thesis text, figures and tables from the author's project results	Every figure and table regenerated from the repository data; all references checked against the original source; text reviewed and revised by the author	Chapters 1–9, appendices

Tool	Purpose	How the output was verified	Sections / artefacts affected
Claude (Anthropic)	Help drafting the multilingual symptom and retreat-goal example sentences (German, Russian, Chinese, Japanese) from the English condition descriptions	Examples checked against the domain expert's condition descriptions; quality measured by leave-one-out accuracy per language (Table 6.3)	`backend/data/*_training_data_*.json`

No other generative AI tools were used. A typical workflow was to describe a task (for example, adding input validation to an API endpoint or drafting a section from the project results), review the suggested code or text, run it against the repository and its tests, and then revise it. All reported metrics and figures are computed from the project repository by the training notebook and test scripts.